

The Effects of Focused Deterrence on Crime and Community Outcomes in Philadelphia

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Abstract

Focused Deterrence is a crime-reduction strategy that combines an increased probability and severity of punishment with direct communication of consequences to individuals identified as likely offenders. Philadelphia implemented Focused Deterrence in 2013 in response to rising gun violence. We estimate this policy's effects on crime rates as well as community downstream effects on rental pricing and housing valuations. Using a standard difference-in-difference design, we find that the number of shooting victims declined by 41.3 percent and gun crime incidents fell by 24.2 percent. Total reported crime incidents declined by roughly 13.8 percent, reflecting reductions in both violent and non-violent offenses. This reduction in crime resulted in increases in rental prices between 6.0 to 7.5 percent, and housing values from 3.6 to 7.0 percent. Together, these findings suggest that Focused Deterrence's crime-reduction effects extend beyond its direct targets, and that this broader improvement in neighborhood safety is capitalized into local housing and rental markets.

Keywords: Crime, Deterrence, Rental Markets, Housing Markets

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1 Introduction

Violent crime imposes substantial costs on both victims and the broader communities where it concentrates, and researchers and policymakers have pursued a wide range of strategies to reduce it. These costs extend well beyond the direct harm to victims, such as medical expenses, lost income, and criminal justice expenditures, to include the broader toll on the communities where violence concentrates. A newer range of strategies has moved beyond traditional, broad-based policing toward more targeted approaches, including hot-spot policing, environmental design interventions such as installing street lighting or restoring vacant lots, and community-led crime prevention efforts (Braga et al. (2019); MacDonald et al. (2026); Nubani et al. (2023)).

One such strategy, Focused Deterrence (FD) takes a more targeted approach: rather than addressing crime broadly across a place, FD aims to change the behavior of high-risk offenders, such as gang members responsible for violent crime in an area, through a coordinated menu of law enforcement, community involvement, and social services actions communicated directly to these offenders through call-in meetings (Braga et al. (2018); Braga and Weisburd (2012); Kennedy et al. (1996)). While FD’s effectiveness at reducing the short-run, targeted violent crime it directly addresses is well studied, less is known about its effects on other, non-targeted criminal offenses and how the crime reductions it produces generate downstream effects in the communities where it operates.

This paper examines three interrelated questions about the effects of Focused Deterrence. First, how effective is FD at reducing violent crime? Second, does FD generate spillover effects on non-violent crime that falls outside its stated targets? Third, to the extent that FD reduces crime, do these reductions capitalize into local housing and rental markets, translating public safety gains into measurable economic gains for treated communities?

Philadelphia adopted this crime reduction approach in 2013 in response to rising gang-related gun violence in the city. Despite FD’s growing adoption in cities around the world, its effectiveness have varied considerably across implementation contexts and less is known about how FD’s implementation impacts the broader communities where it occurs (Braga et al. (2018)). This raises

a direct question: how did Focused Deterrence affect crime rates in Philadelphia, and did any resulting reductions in violence generate downstream effects on local rental and housing markets?

The targeted nature of FD in Philadelphia provides an appropriate context to causally estimate the impact this crime reduction strategy. To estimate the causal effect of Philadelphia’s Focused Deterrence strategy on crime and downstream neighborhood outcomes, we exploit geographic and temporal variation in the policy’s implementation across census block groups in Philadelphia. Because the program was implemented in a defined set of neighborhoods in Southern Philadelphia beginning in April 2013, we construct a matched comparison group using propensity score matching on pre-treatment demographics and crime trends from the pool of other Philadelphia County census block groups. Using this matched sample, we estimate event study and difference-in-differences specifications with block group and year fixed effects to estimate the dynamic and average treatment effects on treated block groups. As robustness exercises, we find similar results with alternative matching specifications and using probability weighting instead of matching.

Using this strategy, we find that the FD policy had a significant impacts on crime and downstream outcomes in treated communities. In our main specification, we find the number of shooting victims declined by 41.35 percent and the total number of gun crime incidents fell by 24.15 percent. Total reported crime incidents declined by approximately 13.8 percent, which included reductions in both violent serious offenses (Part 1 offenses) and non-violent less-serious offenses (Part 2 offenses). We find these crime reductions produced by the FD strategy generated positive downstream effects on treated neighborhoods as reflected in local rental markets and residential property values. Following implementation of the strategy, rental prices increased by approximately 7.0 percent and housing values increased by approximately 6.0 percent. These estimates suggest the crime reduction attributable to the policy were capitalized into local property markets consistent with a broader literature linking changes in crime with changes to prices in rents and home values ([Thaler \(1978\)](#), [Linden and Rockoff \(2008\)](#), [Ihlanfeldt and Mayock \(2010\)](#), [Pope and Pope \(2012\)](#)).

Our findings contribute to the growing literature measuring the effectiveness of Focused Deterrence as a crime-reduction strategy. Meta-analyses by [Braga and Weisburd \(2012\)](#) and [Braga et al. \(2017\)](#) broadly find Focused Deterrence reduces violent crime across a wide array of implementa-

tions, but most individual analyses focus on one outcome in the immediate post-treatment period, such as shooting victims or number of homicides. We contribute to this literature in three main ways. First, our setting allows us to causally identify the impacts of FD on crime, a feature that is missing from many analyses in the Focused Deterrence literature. Second, we expand the set of outcomes considered beyond the direct violent crime measures that exist in the current literature, allowing us to investigate whether FD generates spillovers onto non-violent offense types. Third, our setting allows us to examine the policy’s effects over a longer time horizon of six years, rather than the typical short-run two-year windows of prior evaluations, offering evidence of how Focused Deterrence’s crime-reduction effects evolve over time.

Our results on shooting victims and gun crime incidents align with this broader evidence of FD’s effectiveness at reducing violent crime, while our results on non-violent offenses suggest that Focused Deterrence generates spillover effects onto non-targeted offense types. Moreover, these effects persist over the long run, even after the official end of the policy, suggesting Focused Deterrence can have a lasting impact on crime in treated communities beyond the active intervention period.

More directly, our paper builds on the Philadelphia Focused Deterrence literature pioneered by [Roman et al. \(2019\)](#), extending their analysis with an updated panel, an alternative empirical framework, and downstream economic outcomes not previously examined in this setting. Our paper also speaks to the hedonic-capitalization literature on crime and housing prices ([Thaler, 1978](#); [Linden and Rockoff, 2008](#); [Pope and Pope, 2012](#)) by using a geographically targeted policy intervention to identify how crime reductions capitalize into housing values and rental prices.

The remainder of the paper proceeds as follows. Section 2 describes the Philadelphia Focused Deterrence intervention, the institutional details relevant for interpretation, and the timeline of call-ins and enforcements during the evaluation period. Section 3 reviews the literature on Focused Deterrence policies, and how crime impacts rental and housing markets. Section 4 describes the data sources, construction of the analytical sample, and the propensity-score matching process. Section 5 outlines the empirical strategy used in the main specification. Section 6 presents the main results of FD on crime and downstream outcomes. Section 7 analyzes the sensitivity of our results to robustness checks such as alternative matching specifications. Section 8 discusses

our results in terms of feasibility and magnitudes and measuring the benefits of the policy, while Section 9 concludes.

2 Policy Background

Focused deterrence policies are a type of crime reduction public policy that aims to combat violent crime through changing offender behavior by implementing a coordinated menu of law enforcement, community involvement, and social service actions (Braga et al. (2018); Braga and Weisburd (2012); Kennedy et al. (1996)). These policies are constructed based on the theoretical framework of deterrence, which is premised on the assumption that criminals are rational actors and that crime can be deterred through a change in the costs or benefits of criminal activity (Becker (1968), Nagin (1998), Nagin (2013)). Focused deterrence concentrates the threat of sanction on a small, identifiable population, typically gangs or criminally active street groups, responsible for a disproportionate share of violent activity in a given area, and advertises changes in the costs of violent crime to prevent offenses or re-offenses.

The approach was first developed and implemented in Boston, Massachusetts during the mid-1990s under the name Operation Ceasefire. The intervention served as a coordinated effort between the Boston Police Department, federal and state prosecutors, probation and parole authorities, city youth outreach workers, and researchers from Harvard’s Program in Criminal Justice Policy and Management to reduce youth gun violence, which was rapidly on the rise in the late 1980s and early 1990s (Kennedy et al. (1996)). The resulting intervention, which combined the deterrence approach with direct law-enforcement action against illicit firearm trafficking, saw a 63 percent decline in mean monthly youth homicides following implementation (Braga et al. (2017)). Since the initial pilot program in Boston, the strategy has been implemented in 84 U.S. cities ranging from large metropolitan areas such as Chicago and Los Angeles to smaller cities such as New Haven, Connecticut and Roanoke, Virginia (Braga et al. (2018), Papachristos et al. (2007), Tita et al. (2011), Sierra-Arevalo et al. (2017), Saunders et al. (2015)). The intervention strategy has generally been used to address three different types of criminal behavior including gang and criminally active

group violence reduction, crime driven by street-level drug markets, and preventing repeat offending of high-risk individuals ([Braga et al. \(2017\)](#), [Kennedy et al. \(1996\)](#), [Kennedy \(2012\)](#), [Kennedy and Wong \(2009\)](#), [Deuchar \(2013\)](#), [Papachristos et al. \(2007\)](#)).

These policies are often implemented through a coordinated effort between different law enforcement agencies, legal agencies, city governments, local community figures, and research organizations. Focused deterrence policies begin with interagency information sharing that identifies the hotspots of violent crime and the groups responsible for crime in those areas. Next, a menu of "levers" is bundled across different agencies to serve as a coordinated response to continued violence. These elements are then activated through an initial call-in meeting, in which members of the targeted group are brought in to establish communication between these groups and law enforcement that further violent crime will not be tolerated and to convey the new punishments that will result to all members of the targeted groups in the event of further violence. In the event of further violence after this call-in meeting, enforcements of the advertised menu of punishments are conducted swiftly to demonstrate credibility of the advertised consequences.

The lever bundle deployed in focused deterrence policies typically spans both formal and informal mechanisms of social control. The formal mechanisms operate through three categories of pre-existing legal authority. Prosecutorial levers include higher bail terms, tougher plea bargains, and the use of federal prosecution to escalate the consequences of new offenses committed by targeted individuals. Police enforcement levers include intensified law enforcement activity in target areas, the execution of outstanding warrants against members of the targeted group following a triggering event, and the targeting of illegal sources of income associated with targeted individuals or groups. Probation and parole levers include increased probation and parole requirements for targeted individuals. Beyond these formal sanctions, focused deterrence policies also incorporate community-based mechanisms of informal social control. Social service providers offer voluntary educational, vocational, and treatment programming to individuals who choose to disengage from violence, raising the opportunity cost of continued offending. Community moral voices, including faith leaders, well-respected community groups, and family members of past victims of gun violence, reinforce the anti-violence message by setting clear normative expectations within the af-

affected community. Across implementations, the distinguishing feature of focused deterrence is not the introduction of novel sanctions but the coordinated, swift, and credible deployment of formal authority that already existed across separate agencies, paired with the explicit mobilization of informal social control through community voices and the offer of legitimate alternatives through social services (Kennedy et al. (1996); Roman et al. (2019)).

In focused deterrence policies, the cost-benefit calculus of committing violent crimes is changed along multiple dimensions. On the cost side, three margins are affected: certainty, severity, and swiftness of punishment. First, the probability of getting caught increases through the use of inter-agency intelligence sharing on targeted groups, direct advertisement to targeted groups, community involvement and awareness of the goal of reducing violent crime, and through the introduction of group-level liability, in which any violent act attributable to a gang triggers an enforcement response against all of its members. Second, the severity of punishment of violent crime increases through the use of coordinated law enforcement, legal, and other "lever" bundles, such as execution of outstanding warrants, strict probation and parole conditions, and high-bail or sentencing enhancements. Notably, most of these tools were already legally available, but the innovation of focused deterrence lies in the credible commitment to deploy them swiftly, predictably, and in coordinated fashion across agencies. Third, the swiftness of punishment increases through pre-coordinated enforcement protocols that allow agencies to respond to violent acts within days rather than weeks or months. On the benefit side, the opportunity cost of committing crime is raised through social service offerings, including GED programming, vocational training, and substance abuse treatment, that improve returns to legitimate activity for targeted individuals.

Communication is the binding mechanism that activates these channels: call-in meetings, in which targeted gang members are required to attend and are explicitly informed of the enforcement consequences awaiting them and their group, are designed to update offenders' subjective beliefs about the probability and severity of sanction. Network diffusion of the message through gang affiliations is intended to extend this updating to non-attending members and to potential offenders within the broader community.

Following substantial increases in violent crime in 2006 and again in 2010, the Mayor of Philadel-

phia, in conjunction with the Philadelphia Police Department (PPD) and the Philadelphia District Attorney’s Office (DAO), began reviewing potential violence reduction strategies. Beginning in the fall of 2012, this working group became familiar with the Focused Deterrence strategy developed during Operation Ceasefire and began shaping an analogous intervention for Philadelphia. The working group expanded to include the Adult Probation and Parole Department, Juvenile Probation, the Philadelphia Housing Authority Police Department, social service agencies, researchers from Temple University, and the local federal prosecutor’s office (Roman et al. (2017); Roman et al. (2019)).

Unlike Operation Ceasefire, which was implemented citywide, Philadelphia’s intervention was specifically targeted at South Philadelphia. The working group’s selection of South Philadelphia reflected several factors: an entrenched street gang culture, pre-existing interagency relationships among local law enforcement, manageable geographic coverage, clustered gang territories, and a pre-existing structured intelligence apparatus within PPD’s Central Intelligence Unit (Roman et al. (2017); Roman et al. (2019)).

The intervention officially launched with an initial call-in meeting on April 17, 2013, at which time 16 active gangs became the focus of the program; the first enforcement was conducted three days later in response to a shooting committed by a targeted group (Roman et al. (2019)). Over the first two years of the policy, four call-in meetings and 16 enforcements were carried out, with ten gangs subject to enforcement actions, some on more than one occasion. The program remained active until early 2017.

Philadelphia’s implementation deployed each of the general lever categories described above, prosecutorial, police enforcement, and probation and parole tools, while also introducing unique levers specific to this implementation of FD. First, housing and utilities-related sanctions included termination of public utility services for nonpayment or illegal connections, review of public housing eligibility, and targeted code enforcement. Second, probation and parole escalation tools included increased supervision intensity, execution of outstanding warrants, revocation of probation for those arrested but not yet convicted of a new offense, and enforcement of outstanding child support orders. Third, prosecutorial tools included requests for high bail following any new arrest and advocacy for

longer or more restrictive sentences upon new convictions, with gang task force members providing testimony at all relevant hearings (Roman et al. (2019)). While most of these levers represented coordinated applications of pre-existing legal authority, the housing and utilities-related sanctions constituted a more distinctive Philadelphia-specific extension of the standard focused deterrence toolkit. Beyond these enforcement levers, social services constituted the benefit-side complement of the intervention. Roman et al. (2019) report that social services made initial contact with approximately 112 group members across 14 street groups between April 2013 and March 2015, with roughly 33 percent engaging in some level of services such as GED programs, substance abuse treatment, or vocational training.

3 Literature Review

The proliferation of focused deterrence programs across cities has generated a substantial body of evidence with which to evaluate their effectiveness. Meta-analyses of this literature find that focused deterrence strategies are associated with a statistically significant, moderate overall crime reduction effect. The strongest impacts are observed in programs targeting serious violence by gangs and criminally active groups, while strategies aimed at repeat offenders more broadly or at drug market crime tend to produce more modest reductions (Braga and Weisburd (2012); Braga et al. (2018)).

The existing evaluation literature on Philadelphia’s Focused Deterrence intervention originates with Roman et al. (2019), who assess the community- and gang-level impacts of the intervention using a propensity score matching design at the census block group level — the same general approach we adopt here. Matching treated neighborhoods in South Philadelphia to comparable untreated neighborhoods elsewhere in the city on baseline violence trends and demographic characteristics, they find a statistically significant 35 percent reduction in criminal shooting rates in treated areas over the 24 months following the April 2013 call-in. The gang-level findings, however, are more mixed: while some models show significant reductions in shootings around the territories of targeted gangs, the majority of specifications do not yield statistically significant gang-level effects, suggest-

ing that the community-level deterrence effect may operate through general deterrence mechanisms rather than through behavioral change among specifically targeted groups. [Roman et al. \(2020\)](#) complement this analysis by examining the enforcement levers deployed during the intervention and find no evidence of indiscriminate increases in arrests or shifts in the composition of crime types in treated areas, suggesting that the observed shooting reductions were not an artifact of broader enforcement escalation. Subsequent work has examined heterogeneity in the intervention’s effectiveness across targeted gangs, finding that variation in group-level social media activity partly explains differential responses to the deterrence message, and has documented the spatiotemporal patterns of shooting hot spots before, during, and after implementation ([Hyatt et al. \(2021\)](#), [Boschan and Roman \(2024\)](#)).

Crime is best understood in this literature as a localized disamenity that is capitalized into housing markets, but the most persuasive empirical papers move well beyond simple hedonic correlations. [Thaler \(1978\)](#) provides the basic background idea that households are willing to pay for crime control, yet later studies rely on designs that more directly address the endogeneity of crime and neighborhood change. In London, [Gibbons \(2004\)](#) is summarized as finding roughly a 10 percent decline in property values for a one-standard-deviation increase in local property crime. U.S. quasi-experimental evidence points in the same direction: [Linden and Rockoff \(2008\)](#) show that houses within 0.1 miles of a registered sex offender sell for about 4 percent less, while [Pope \(2008\)](#) similarly finds localized housing-market penalties following sex-offender registry disclosures. At the broader neighborhood level, [Ihlanfeldt and Mayock \(2010\)](#) use a nine-year Miami-Dade panel with first-difference IV methods and show that robbery and aggravated assault, rather than all crime categories, are the crimes that materially depress housing values; in their preferred specifications, a 1 percent increase in neighborhood crime density lowers housing prices by roughly 0.1 to 0.3 percent. Related work exploiting the 1990s crime decline is summarized as estimating crime elasticities of property values between about -0.15 and -0.35 [Pope and Pope \(2012\)](#). Taken together, this literature strongly suggests that public safety is priced into local housing demand, with the clearest evidence showing up in sale prices and housing values, even if direct rent evidence is thinner in the core causal papers.

While the existing literature has focused primarily on reductions in criminal shooting victims and the mechanisms behind those declines, this paper examines the broader impacts of the intervention. We first investigate whether the policy generated reductions in non-targeted crime categories, including nonviolent and property crime, alongside its effects on targeted violent crime. Second, we examine the downstream economic consequences of the intervention by estimating its impact on rental prices and housing prices in treated communities. Because the documented community-level deterrence effect operates on the local amenity bundle rather than on a small set of identified offenders, neighborhood-level economic outcomes are the natural object of study for assessing the broader impacts of the intervention.

4 Data

4.1 Data Sources

To analyze the impact of Philadelphia’s Focused Deterrence Intervention on crime and downstream effects on treated communities, we construct a census block group by year panel of Philadelphia County spanning from 2006 to 2019. The census block groups in our data are defined using the 2010 census block group definitions with crosswalks for pre-2010 block groups completed using IPUMS’s National Historical Geographic Information System (NHGIS) block group crosswalks ([Schroeder et al. \(2025\)](#)). Treatment assignment data at the block group level were collected from [Roman et al. \(2017\)](#) with the policy beginning in April 2013 and ending in late 2017. We end our sample in 2019 to avoid confounding issues from the COVID-19 pandemic, which produced well-documented disruptions in both crime patterns and policing behavior ([Regalado et al. \(2022\)](#)).

Crime is measured through two different categories of outcomes: the number of incidents and the number of shooting victims. A crime incident is a reported offense made to the police in which the offense is committed by the same offender, or group of offenders acting in concert, at the same time and place ([Federal Bureau of Investigation \(2019b\)](#)). Shooting victim data are recorded separately, with one observation per victim, as a single shooting incident may involve multiple victims. Crime incident and shooting victim data were collected from the Philadelphia Police Department’s (PPD)

Crime Incidents and Shooting Victims datasets, available through OpenDataPhilly ([Philadelphia Police Department \(2016a\)](#); [Philadelphia Police Department \(2016b\)](#)). Crime incident data are available for all of Philadelphia County at an incident-by-address level over the full sample period (2006-2019), while shooting victim data are available at the victim-by-address level beginning in 2010. These data are then aggregated by crime incident and shootings categories, census block group, and year level.

To examine the downstream effects of the policy on community-level outcomes, we gather contract rent and housing value data from the American Community survey (ACS) 5-year estimates at the census block group level from 2009 to 2019 ([U.S. Census Bureau \(2019\)](#)). We acknowledge that the year-to-year variation in ACS-based outcomes may be attenuated given the rolling nature of the five-year estimates; however, this dataset allows us to view these crucial outcomes at a granular level.

We draw on the same ACS 5-year estimates to construct demographic and socioeconomic information at the census block group level, including measures of population, income, educational attainment, and racial composition, which we use both as matching covariates in the propensity score matching procedure, and as controls in estimation of the policy’s treatment effect.

We have several different outcomes of interest. First, we investigate the number of shooting victims within a census block group within each year. Second, we investigate how Focused Deterrence impacted the total number of crimes in a census block group each year. We then break down these crimes by category based on the Federal Bureau of Investigation (FBI) Uniform Crime Reporting categories ([Federal Bureau of Investigation \(2019b\)](#)). These categories include part one offenses, which include serious crimes against people and property — homicide, rape, robbery, aggravated assault, and arson — and part two offenses, which include other crimes against persons (assaults of a non-aggravated nature), public disorder crimes (vandalism, prostitution, drunkenness, etc.), drug and alcohol offenses, white collar crimes (fraud, embezzlement, forgery), and all other crimes (weapons violations, possessing stolen property, state or local laws not otherwise specifically identified).¹ We also construct a crime measure, the number of crimes involving a firearm, which is the

¹A full listing of FBI UCR offense categories and their definitions is available in [Federal Bureau of Investigation](#)

total number of aggravated assaults with a firearm, robberies with a firearm, weapons violations, and homicides. The crime incident dataset does not specify the weapon used for homicides; however, the FBI reports that around 70 percent of homicides are committed with a firearm ([Federal Bureau of Investigation \(2019a\)](#)). Regardless, this measure might overstate the count of gun crime incidents. As a robustness check, we adjust the definition of gun crimes in several ways.

Third, we investigate the downstream effects of crime reduction on rental and housing markets in treated communities. Our primary outcomes of interest are contract rent and housing structure value, both measured in 2019 real dollars. For each, we examine impacts at multiple points of the distribution — lower quartile, median, and upper quartile. We expect heterogeneous effects across the distribution: because crime operates as a disamenity that is capitalized into rents and prices, units in high-crime areas should trade at a discount relative to otherwise comparable units. A reduction in crime should therefore generate the largest proportional gains at the lower end of the rent and value distributions, where the pre-treatment crime discount was largest.

4.2 Summary Statistics

Figure 1 shows the number of criminal homicide incidents by census block group for Philadelphia county in 2011, 2-years prior to the implementation of the Focused Deterrence intervention. As mentioned in the policy background, Southern Philadelphia was chosen for intervention based on entrenched street gang culture, pre-existing interagency relationships among local law enforcement, manageable geographic coverage, and clustered gang territories ([Roman et al. \(2017\)](#); [Roman et al. \(2019\)](#)). As seen by the figure, several census block groups in Southern Philadelphia are "hot spots" for criminal homicides as these several communities experienced between 1 to 6 homicides in 2011. However, homicides in Philadelphia county are not located exclusively in Southern Philadelphia in frequency or severity as both Western Philadelphia and Northern Philadelphia have more census block groups that experienced homicides, and census block groups that experienced large severity in the number of homicides. This geographic dispersion of homicides and shootings around Philadelphia helps motivate our propensity score matching technique, as we will be able to match similar

census block groups in Southern Philadelphia to other census block groups around Philadelphia in terms of crime incident, shooting victim, and demographic trends.

Table 1 compares the pre-treatment means of shootings victims, crime incidents, and community characteristics of South Philadelphia to the rest of the city. Compared to the rest of Philadelphia, Southern Philadelphia has statistically fewer shooting victims, Part 1 crime incidents, Part 2 crime incidents, and gun crime incidents. In addition, Southern Philadelphia census blocks have lower Black and Hispanic population shares, a higher share of the population with a Bachelor’s degree or above. These census blocks also have a higher labor force participation rate, lower unemployment rate, smaller shares of households on SNAP benefits and below the poverty line, and higher median household income. These differences in shooting victims, crime incidents, and community characteristics once again highlights the necessity of identifying a comparable comparison group to Southern Philadelphia, as these communities look noticeably different in observables on average from the rest of the city.

Table ?? compares pre-treatment means of contract rents and housing values in South Philadelphia to the rest of the city. Lower quartile rents are similar across the two areas, but the distributions diverge above that point: median and upper quartile rents in South Philadelphia exceed those in the rest of the city by approximately 25and80, respectively. Housing values are significantly higher across the entire distribution, with lower quartile, median, and upper quartile values in South Philadelphia exceeding the rest of the city by roughly \$37,000, \$ 64,000, and \$ 94,000, respectively. As with the demographic and crime comparisons above, these level differences in housing and rental markets motivate our propensity score matching approach over naive comparisons to the rest of the city.

In addition to the pre-treatment averages of these characteristics being noticeably different, the trends in the shootings and crime incidents during the pre-treatment years are also different. Figure 2a shows the trend in the average number of shooting victims in Southern Philadelphia compared to the rest of Philadelphia. While both areas were experiencing around an average of 1.2 shooting victims in a census block in 2010, Southern Philadelphia was already experiencing a noticeable decline in the average number of shooting victims leading up to the intervention in 2013 that was

not seen in the rest of the city.

Figure 2 shows the trends in crime incidents in Southern Philadelphia compared to the rest of the city broken down by total, Part 1, Part 2, and gun crime incidents. Once again there are noticeable differences in the trends of crime incidents during the pre-treatment period. While the trends in Part 1 crime incidents look comparable across the two groups, Southern Philadelphia was experiencing a downward trend in the average number of total crime incidents, Part 2 crime incidents, and gun crime incidents not seen in the rest of the city as early as 2010. Therefore, a naive comparison utilizing all other Philadelphia census block groups to Southern Philadelphia in a causal analysis would struggle to pass the parallel trends assumption necessary to calculate an accurate treatment effect of Focused Deterrence.

Figures 3 and 4 show trends in contract rents and housing values in South Philadelphia and the rest of the city, broken down by lower quartile, median, and upper quartile. In contrast to the crime trends, which diverge between the two areas, pre-treatment rental prices and housing values appear visually comparable. Both series show a sharp decline from 2009 to 2010 reflecting the post-Great Recession housing market correction, followed by a slow recovery. Beginning around 2013, South Philadelphia experiences sharper growth in rents across the distribution; a similar divergence in housing values emerges around 2015, consistent with the slower-moving housing market responding with a lag relative to rents. While this post-treatment divergence is suggestive, attributing it to focused deterrence requires isolating a control group with comparable pre-treatment crime trends, which our matching procedure constructs.

4.3 Propensity Score Matching

To begin the process of finding a causal treatment effect of Focused Deterrence on the number of shooting victims, the number of crime incidents, and downstream effects, we utilize propensity score matching. Specifically, we implement a nearest-neighbor matching technique with replacement which identifies census block groups among the potential pool of communities from the rest of the city based on the estimated probability of treatment selection. We estimate the probability of treatment selection with the following equation:

$$Pr(Treat_i) = \Lambda(\alpha + \beta_1 \Delta Part1_i + \beta_2 \Delta Part2_i + \beta_3 \Delta Gun_i + \gamma \bar{X}_i + \epsilon_i) \quad (1)$$

where $Pr(Treat)$ is the probability of census block group i ever receives treatment. This probability was estimated using a logistic regression as denoted by Λ . The covariates used to predict the probability of treatment were selected based on justification of treatment selection as outlined in [Roman et al. \(2019\)](#). First, $\Delta Part1_i$, $\Delta Part2_i$, and ΔGun_i represent the trend (first difference) in Part 1, Part 2, and gun crime incidents from 2006 to 2012 in census block group i . Second, $\bar{X}$ represent a vector of community characteristics including youth male density (share of population that is male between the ages of 14 and 24), total population, Black population share, Hispanic population share, and economic disadvantage index (created using a census block group's unemployment rate, share of households below the poverty line, and share of households on SNAP benefits). These community characteristics used as covariates in this matching process are averages of these values between the years of 2010 to 2012. The years utilized in the creation of the crime trends, and community characteristic averages were selected based on of data availability in from PPD and ACS. Finally, census block groups located within the 22nd Police District and several block groups within the 30th Police District were not allowed to be matched on due to their exposure to Philadelphia CeaseFire, a coinciding shooting reduction policy happening in a small area in central Philadelphia ([Roman et al. \(2018\)](#)).

Table 3 and Figure 5 show the comparison between the unmatched and matched samples as a result of the above matching procedure. The unmatched sample shows substantial and statistically significant imbalance across nearly every covariate, particularly in Part 2 crime trends, percent Black, and percent Hispanic. Southern Philadelphia had steeper declining crime trends, were less Black and less Hispanic compared to the broader pool of the rest of Philadelphia. From the 1,182 possible comparison block groups, the matching process found 119 comparable census block groups and substantially reduced bias across almost every covariate. The one caveat is that after matching, percent Black still has a statistical difference in means between the treated and comparison group. In the matched sample, census block groups in Southern Philadelphia statistically have a higher

black population share compared to the match comparison group.

Propensity score matching techniques rely on two assumptions - conditional independence and common support. While conditional independence is not inherently testable, we know that South Philadelphia was selected for treatment based on the observable characteristics of entrenched street gang culture, pre-existing interagency relationships among local law enforcement, manageable geographic coverage, and clustered gang territories (Roman et al. (2017); Roman et al. (2019)). Covariate selection was based on capturing these observable determinants of selection. Common support is directly testable. Figure 6 shows the kernel density of propensity scores for the entire sample of census block groups in Philadelphia. Census block groups in South Philadelphia and the rest of the city have possible range of propensity scores ranging from 0 to 1, providing evidence of common support. Of note, there is some bunching of propensity scores in the rest of Philadelphia that is near zero, speaking to the fact that many census block groups in Philadelphia are not similar to the treated units in Southern Philadelphia. Figure 7 shows another kernel density of propensity scores in the post-matched sample. There is near perfect overlap of propensity scores among both the treated and control samples across the possible ranges of propensity scores from 0 to 0.85.

Visually, the resulting treated and control groups can be seen on the map of Philadelphia in Figure 8. The matched control group selected census block groups from different areas around Philadelphia predominantly in western, northwestern, and northeastern Philadelphia. Only a few census block groups adjacent to the Southern Philadelphia treatment border were selected as control groups, alleviating some concerns of direct crime spillover effects of the intervention into control block groups. For robustness, we remove these control bordering block groups.

Figures 9 show the timeseries of the average shooting victims, and crime incidents among the treated and control groups in the matched sample. Compared to the pre-matched sample (with the comparison of just the rest of Philadelphia) the trend in shooting victims is similar across both groups prior to the intervention. Following the start of the policy in 2013, there is a noticeable decline in the average number of shooting victims for the first two years of the intervention not seen in the control group. After 2014, the treated areas experience a rise in average shooting victims and then follow a similar trend to control group for the rest of the sample.

In terms of average crime incidents, we now witness a more comparable pre-policy trend in average total number of crime incidents, Part 1, Part 2, and gun crime incidents in the matched sample. The average total number of crime incidents experiences a drop in the years following the implementation of the FD intervention not seen in the control group, with this drop being primarily driven by Part 2 crime incidents. While Part 1 crime incidents follow similar trends post-implementation, we witness a spike in Part 1 Crime incidents in the control group during 2016 in the control group that is not seen in the treated group which might speak to the policy having an effect of maintaining a downward trend in particularly violent crimes during a period of growing violence in the control group. The trend in the average gun crime incidents declines in the first two years of the policy in the treated block groups compared to the control block groups. Following 2014, the average number of gun crime incidents trend in a similar fashion among both groups for the rest of the period.

Table 4 presents the raw means of the outcomes of interest for the control and treated group in the pre-intervention period and post-intervention period. Comparing the change in the average number of shooting victims by group, the control group experienced an increase of shooting victims from an average of 0.179 shooting victims in a census block group per year, an increase of 164.24 percent. Meanwhile, the treated group demonstrated an increase in the average number of shooting victims from 0.387 to 0.592, an increase of 52.97 percent. Although the number of shooting victims increased for both groups, both groups experienced declines in the average total number of crime incidents, and within each category of crime. South Philadelphia experienced larger reductions in crime with a 31.67 percent decline in average total crime incidents, 22.25 percent decline in Part 1 crime incidents, 36.94 percent decline in Part 2 crime incidents, and 41.37 percent decline in the average number of gun crime incidents. Meanwhile, the control group experienced a 15.95 percent decline in average total number crime incidents, 16.05 percent decline in Part 1 crime incidents, 15.86 percent decline in Part 2 crime incidents, and 20.62 percent decline in gun crime incidents respectively.

While the number of shooting victims did not experience a decline in the post-period, the treated census block groups experienced a smaller increase in magnitude in the average number of shooting

victims compared to the control group. In addition, the treated census block groups experienced larger declines in all crime incidents compared to the control block groups suggesting a positive crime reduction effect of the intervention. In the next section, we outline our empirical strategy to determine how much of the trend in crime reduction was attributed to the Focused Deterrence intervention.

Table 5 presents the post-matching raw means of the rental and housing value outcomes for the control and treated groups in the pre-intervention and post-intervention periods. Beginning with rents, the control group experienced modest growth across the distribution: lower quartile rents rose from \$726.01 to \$771.45, an increase of 6.26 percent; median rents rose from \$913.24 to \$957.06, an increase of 4.80 percent; and upper quartile rents rose from \$1,123.30 to \$1,186.40, an increase of 5.62 percent. The treated group experienced substantially larger growth across the same points of the distribution, with lower quartile rents rising from \$651.93 to \$740.14 (13.53 percent), median rents rising from \$835.71 to \$948.56 (13.50 percent), and upper quartile rents rising from \$1,065.60 to \$1,208.16 (13.38 percent).

Turning to housing values, both groups experienced declines at the median and upper quartile, consistent with the broader housing market correction; however, the treated group experienced smaller declines, and at the lower quartile experienced growth. In the control group, lower quartile housing values declined from \$176,051 to \$173,553 (1.42 percent), median housing values declined from \$250,280 to \$228,649 (8.64 percent), and upper quartile housing values declined from \$336,164 to \$311,539 (7.32 percent). In the treated group, lower quartile housing values rose from \$169,537 to \$177,459 (4.67 percent), median housing values declined only modestly from \$246,398 to \$242,500 (1.58 percent), and upper quartile housing values declined from \$338,636 to \$326,019 (3.73 percent).

Across all six outcomes, the treated group outperformed the control group in the post-intervention period. The treatment-control gap in rent growth is approximately uniform across the distribution, at roughly 7 to 9 percentage points. The treatment-control gap in housing values is more concentrated in the lower and middle of the distribution—approximately 6 percentage points at the lower quartile and 7 percentage points at the median, compared to roughly 4 percentage points at the upper quartile—broadly consistent with the prediction that crime reduction should disproportionately

benefit the lower end of the housing market, where the pre-treatment crime discount was largest. In the next section, we outline our empirical strategy to determine how much of these differential trends can be attributed to the Focused Deterrence intervention.

5 Empirical Strategy

5.1 Event Study

To investigate the dynamic treatment effects and pre-treatment trends of the Focused Deterrence policy, we estimate the following equation:

$$y_{i,t} = \sum_k \beta_k D_{i,t}^k + \alpha_i + \delta_t + \lambda X_{i,t} + \epsilon_{i,t} \quad (2)$$

where $y_{i,t}$ is either the number of shooting victims, number of crime incidents, or downstream effect outcome of interest in census block group i in year t . $D_{i,t}^k$ is an indicator of whether census block group i is in its k -th year relative to treatment at time t . The reference year is one year prior to the policy, 2012. α_i and δ_t represent census block group and year fixed effects respectively. Finally, $X_{i,t}$ is a vector of controls including demographics. Standard errors are clustered at the census block group level.

5.2 Difference-in-Differences

To investigate the static average treatment on treated (ATT) effect of the Focused Deterrence policy, we estimate the following equation:

$$y_{i,t} = \beta(Post \times Treat)_{i,t} + \alpha_i + \delta_t + \lambda X_{i,t} + \epsilon_{i,t} \quad (3)$$

where $y_{i,t}$ is either the number of shooting victims, number of crime incidents, or downstream effect outcome of interest in census block group i in year t . $(Post_t \times Treat_i)$ is an interaction indicator of $Post$ which is equal to 1 if the year is equal to or after 2013, and $Treat$ is equal to

1 if the census block group ever received the Focused Deterrence Treatment. α_i and δ_t represent census block group and year fixed effects respectively. Finally, $X_{i,t}$ is a vector of controls including demographics. Standard errors are clustered at the census block group level.

In this specification, β represents the causal effect of Focused Deterrence on the shooting, crime, or downstream outcome of interest. Causal identification of this treatment effect relies on parallel trends of treatment and control groups, stable unit treatment value assumption (SUTVA), no pre-treatment anticipation, and composition stability. While parallel trends is not directly testable, the sample selection completed through propensity score matching has resulted in comparable census block groups in crime trends and demographics. Investigation of raw means pre- and post-matching reveals better comparability in crime trends prior to the policy. In addition, event studies allow us to investigate pre-treatment differences between our treated and control groups.

The stable unit treatment value assumption in this context would mean that criminal behavior outside of Southern Philadelphia is not impacted by Focused Deterrence — either through crime reduction or through crime displacement to other parts of the city. The nature of the shootings predominately attributed to gang violence tends to be extremely localized in nature around neighboring gang border territories (Roman et al. (2017)). As the nature of gang territories tends not to relocate, it is unlikely that the shooting incidents targeted by the policy would be relocated to areas outside of bordering gang territories. In addition, there are several geographical barriers that would deter spillovers to other parts of the city from Southern Philadelphia. As seen by Figure 8 Southern Philadelphia is separated from Northern Philadelphia via Interstate 676 that cuts through the center of Philadelphia. In addition, Southern Philadelphia is separated from Western Philadelphia via the Schuylkill River. These separations help alleviate some concerns about direct geographic spillovers. In addition, for robustness we analyze our results adjusting our comparison group by removing any adjacent census block groups to Southern Philadelphia.

As outlined by the policy background, the working group responsible for implementing the Focused Deterrence policy decided on implementing a FD model and the coordination effort between all the necessary jurisdictions wasn't achieved until late 2012 (Roman et al. (2017, 2019)). The first call in meeting was held shortly after in April 2013 which started the implementation of the policy.

This brief recap of the timeline of implementation highlights the fact that it is unlikely that crime activity, primarily gang crime activity, experienced any anticipatory behavior changes prior to the implementation of the policy in April 2013.

Treated and comparison groups are stable over the course of our sample as once census block groups are treated in 2013 they remain treated until the end of the intervention in 2017. Control census block groups do not experience exposure to Focused Deterrence or any other known crime reduction policies over the course of the sample. With these assumptions in mind, the next section shows the results of both specifications.

6 Results

6.1 Shootings and Crime Results

First, we present the results of the event study specification on the number of shooting victims and crime incidents. Figure 12a presents the event study results for the number of shooting victims from 2010 to 2019. Prior to the implementation of the FD policy in 2013, the trends between the treated and control group are comparable. During the first two years of the intervention there is a decline in the average number of shooting victims by about 0.4 to 0.5 victims per year in the treated census block groups compared to the control groups; however, these estimates are noisy and not statistically significant. Following 2014, the two groups start to return to trending in a similar fashion but never return to pre-treatment differences. The policy officially ended in 2017, and the differences in trends of shootings appear to persist two years following the end of the policy.

Figure 12 shows the event study results for the number of crime incidents from 2006 to 2019. For most of the crime incident categories — total, Part 1, and Part 2 — the trends between South Philadelphia and the matched control group are similar. The trend in gun crime incidents appears to be parallel between the two groups from 2009 onwards, but prior to that the trend in gun crime incidents was higher in Southern Philadelphia compared to the matched control group. Following the implementation of the FD intervention, there is a noticeable drop in the number of total crime incidents in all years of around 20 crime incidents on average in the treated census block group

compared to the control. This drop in total crime incidents is attributed to drops in Part 2 Crime incidents, while Part 1 Crime incidents show potential increases in crime incidents but are not statistically significant for any year. In addition, although Part 1 Crime incidents in aggregate don't appear to be impacted by the policy, the number of crime incidents does show a statistically significant decline post-implementation of the policy from 2014 to 2019.

Table 6 shows the results of the two-way fixed effects difference-in-differences specification. For this specification, we calculate the average treatment effect on the treated for shooting victims from 2010 to 2017, and for crime incidents from 2006 to 2017. This is to capture the strict treatment effect during the policy's duration, and not any reversion of these trends after the policy was discontinued. Starting with the average number of shooting victims per census block group per year, we find that the FD intervention is attributed to a decline of 0.373 shooting victims compared to the control group. Compared to the pre-treatment period mean, this is a reduction of 41.35 percent.

For crime incidents, the FD intervention is associated with a decline of 19.165 total crime incidents on average in treated census block groups compared to control groups. This reduction represents a 13.84 percent decline in total number of crime incidents compared to the pre-treatment mean. Breaking this reduction down by crime incident type, Part 1 crime incidents increased by 0.958 incidents on average per year in treated census block groups compared to the control group; however, this increase is not statistically significant. On the other hand, Part 2 crime incidents demonstrated a 20.123 decrease on average in treated census block groups compared to control block groups. This reduction represents a 22.66 percent decline relative to the pre-treatment Part 2 crime incident mean.

Focusing on gun crime incidents specifically, the Focused Deterrence policy is associated with a decline of 1.298 gun crime incidents and is statistically significant at the 1 percent level. Compared to the pre-treatment mean, this decline represents a 24.16 percent reduction in the number of gun crime incidents in treated census block groups compared to control block groups. These results combined with the reduction in shooting victims would suggest that the intervention had a noticeable impact on reducing shooting activity.

Tables 7 and 8 break down Part 1 and Part 2 crime categories into further subcategories. Part 1

crimes are defined as serious offenses against persons or property and are broken down into violent crimes (serious crimes against persons) and property crimes (serious crimes against property). While we see a slightly positive, statistically insignificant effect on Part 1 crimes overall, violent crimes experience a statistically significant decline of 1.859 incidents on average per year, or around 14.2 percent relative to the pre-treatment mean, in treated census block groups compared to the matched control block groups. Property crimes, on the other hand, experienced a statistically insignificant and imprecise increase over the same period. Overall, this suggests that the FD policy had a significant impact on reducing violent crimes in treated census block groups but had no measurable impact on property crime incidents.

Part 2 crimes are generally less serious or nonviolent offenses and have been broken down into the following subcategories: other crimes against persons, public disorder crimes, drug and alcohol violations, white collar crimes, and a catchall other crimes subcategory. The overall decline in Part 2 crimes is primarily driven by large reductions in public disorder crimes (19.2 percent), drug and alcohol crimes (31.9 percent), and other crimes (32.8 percent). The reductions in Part 2 crime incidents, which drive a large portion of the decline in total crime incidents, suggest that the intervention contributed not only to reductions in violent and shooting crimes, but also deterred a number of nonviolent offenses.

6.2 Downstream Effects

Next, we present the results of the downstream effects of Focused Deterrence as measured through rental prices and housing values via ACS 5-year estimates. As a reminder, both the rental prices and housing valuations represent 5-year rolling average values and are subject to attenuation bias. We anticipate the magnitude of these effects to grow over time as the share of post-policy years in each rolling window increases, and as long-run mechanisms, such as rental price stickiness and gradual information updating among renters and buyers, fully materialize. Therefore, we consider these effects to be a lower bound for the potential effects of Focused Deterrence on these downstream outcomes.

First, Figure 13 presents the event study specification results for lower quartile rents, median

rents, and upper quartile rents. For lower and upper quartile rents, the rental price trends are comparable, as both groups experienced similar declines following the recession in 2009, 2010, and 2011. Following the implementation of the Focused Deterrence policy, both lower and upper quartile rents grew in the treated census block groups compared to the control block groups. Both of these groups experienced growth one year after the policy’s implementation, in 2014, and then became statistically different from the control group from 2015 until the end of the sample. The trend in median rental prices between the two groups is not comparable in the pre-treatment period in 2011, but follows a similar rental price growth trajectory to the other parts of the rental distribution following the implementation of the policy in 2013.

Figure 14 presents the event study specification results for lower quartile, median, and upper quartile housing valuations. Across all three outcomes, the trend in housing valuations is comparable between the treated and control group. Following the implementation of the Focused Deterrence policy in 2013, we see growth among lower quartile and median housing valuations. Lower quartile and median housing valuations increased starting two years after the implementation of the policy in 2015, and the difference generally continued to grow through the end of the sample. Lower quartile housing valuations experienced a sharper increase following the implementation of the policy compared to the two other areas of the distribution. Upper quartile housing valuations appear to have no statistically significant difference between the treatment and control group in the immediate years following the implementation of the Focused Deterrence policy. Upper quartile housing valuations between the two groups began to diverge at the end of the sample in 2018 and 2019 following the end of the policy.

Table 9 shows the results of the two-way fixed effects difference-in-differences specification for the downstream effects on rental prices and housing valuations. Unlike for the crime and shooting results, we calculate the average treatment effect on rental prices and housing valuations using the 2009–2019 sample. This is done to capture the full treatment effect of the policy adjusting for any delayed downstream effects due to potential stickiness in the rental/housing market, or for the diffusion of information about declining crime in the census block groups.

The first three columns of Table 9 show the average treatment effect on the treated of the policy

on rental prices across the lower quartile, median, and upper quartile distribution areas. We find statistically significant increases in rental prices following the implementation of the FD policy of \$41.60 among lower quartile rental price units, \$64.31 among median rental price units, and \$83.33 among upper quartile rental price units, compared to the control census block groups. These increases represent a rental increase of 6.07 percent among lower quartile rental price units, 7.39 percent among median rental units, and 7.64 percent among upper quartile rental units, respectively.

Columns 4 – 6 present the average treatment effect on the treated of the policy on housing valuations across the lower quartile, median, and upper quartile distribution areas. Unlike for the rental prices, we fail to find statistically significant differences between the treated and control groups at the 5 percent level for lower quartile and high quartile housing valuations, though lower quartile housing valuations approach conventional thresholds with a p-value of 0.082. We find positive effects of the policy on housing valuations across the distribution. Lower quartile housing valuations increased by \$11,480, or 6.66 percent compared to the pre-treatment mean. Median housing valuations increased by \$17,384, or 7.01 percent compared to the pre-treatment mean. Finally, upper quartile housing valuations increased by \$12,041, or 3.57 percent compared to the pre-treatment mean.

7 Robustness

7.1 Alternative Matching and Weighting Procedures

The main specification matched treated census block groups in Southern Philadelphia to other census block groups from around Philadelphia based on pre-policy demographics and long-run crime trend variables. The justification for this comparison group was to compare treated census block groups to block groups around Philadelphia that were experiencing similar trends in gun crime incidents, severe part 1 violent crime incidents, and less severe part 2 crime incidents. As a robustness exercise, we follow a similar matching procedure utilizing alternative matching criteria, such as matching pre-policy crime levels (averages), and matching crime levels and trends to test the sensitivity of our results to the matching procedure. In addition, we also test the alternative

to the matching procedure by utilizing augmented inverse probability weights which weighs the full sample of all Philadelphia census block groups based on their probability of receiving treatment based on the same controls in our main specification (i.e. demographics and long-run crime trends).

Table 10 shows the results of this robustness specification exercise across our crime dependent variables of interest. Across all four specifications, the main findings on shootings and crime remain stable and comparable. The intervention continues to be associated with a statistically significant reductions in shooting victims, total crime incidents, gun crime incidents, and total crime incidents, with sizeable effects, albeit varying in size, maintain significance and direction. Shooting victims reductions range from approximately 29% to 41% across specifications, and gun crime incidents range roughly from 11% to 25%. The magnitude of the results decreases with more demanding and restrictive matching procedures which is to be expected. The AIPW weighting procedure also provides comparable results to the mainline specification, providing slightly smaller magnitudes across all categories, but once again is expected given the introduction of census block groups who look non-similar to the treated area.

Table 11 presents the results of these alternative matching and weighting specifications on the downstream results of rental prices and housing markets. For rental price outcomes, the results are reasonably consistent in terms of magnitude and statistical significance. Panel C, which presents the most restrictive matching procedure for matching based on levels and trends, provides smaller and statically noisier estimates of rental prices which is to be expected. Housing value effects are positive in direction but statistically noisier, failing to achieve consistent significance across comparison group constructions and should therefore be interpreted with caution. Taken together Table 10 and 11 suggest that the main findings are not due to a particular comparison group construction. While the crime and downstream effect magnitudes vary predictably with matching or weighting strategy, the direction and significance of the main specification results hold.

8 Discussion

To summarize, we find that Philadelphia’s Focused Deterrence policy produced a significant reduction in the number of shooting victims and gun crime incidents in treated census block groups compared to matched census block groups from the rest of the city, based on demographics and long-run crime trends. In addition to gun-related violence, there was a sizable reduction in total crime incidents driven by a reduction in Part 2 crime incidents (less serious, non-violent crimes) rather than reductions in Part 1 crime incidents (violent crimes against persons and property). We further find evidence that this reduction in crime is capitalized into rental prices and housing valuations, as rental prices increase by around 6.0 to 7.6 percent and housing values increase by 3.6 to 7.0 percent post-policy compared to the control census block groups. Overall, these findings suggest that Philadelphia’s Focused Deterrence policy had a noticeable impact on crime reduction—both in terms of shootings and gun crime incidents and non-violent crime—in treated census block communities, and that rental and housing markets in these communities internalized the reduction in crime into their prices.

The first question one may ask is how these findings, and their magnitude, align with the broader literature on Focused Deterrence. [Roman et al. \(2019\)](#) finds that the number of shooting victims per 10,000 residents declines by 35 percent in the first 24 months following implementation of Philadelphia’s Focused Deterrence policy at the census block group level. Our primary specification, which utilizes a different control group, a varied difference-in-differences specification, and a longer time horizon, finds a 41.4 percent reduction in shooting victims. Our alternative specifications yield a range of reductions from 29.1 to 41.4 percent, which aligns closely with the result found in [Roman et al. \(2019\)](#).

While crime incident counts and composition have not previously been explored in the Philadelphia context, the broader Focused Deterrence literature provides a reasonable benchmark for possible outcomes. Papers that have explored crime incidents as an outcome and found significant effects report reductions in total crime incidents ranging from approximately 15 to 30 percent, and reductions in non-violent offense incidents of around 22 percent ([Saunders et al., 2016](#); [Corsaro](#)

et al., 2013). Our results find a reduction in total crime incidents of approximately 14.0 percent in the main specification, ranging from 7.0 to 20.8 percent across alternative specifications, and a reduction in non-violent incidents (Part 2 incidents) of 22.7 percent, ranging from 9.1 to 31.4 percent across alternative specifications. These results again fall reasonably within the range of crime reductions documented in the existing Focused Deterrence literature.

Our estimates allow for a discussion of the partial benefits of the Focused Deterrence policy. A complete cost-benefit analysis would require more direct estimates for both program benefits and implementation and operational costs. While precise cost data for Philadelphia’s Focused Deterrence program is not currently publicly available, the literature on this program and FD programs more generally suggest that the primary costs of the program consisted of a small number of staff salaries, specifically a community manager and social service manager, and coordination costs between different agencies involved in the program, suggestive of relatively low costs of implementation (Roman et al. (2019), Braga et al. (2017)). Given the absence of a precise total cost figure, we focus on our discussion of the potential benefits of the policy. We provide back-of-the-envelope, conservative lower-bound estimates of program benefits through two complementary frameworks: a cost-of-crime approach and a revealed preference framework using rental and housing market outcomes.

We begin with the cost-of-crime approach, which translates our estimated crime reductions into avoided societal costs using per-incident cost estimates from McCollister et al. (2010), who estimate the societal costs of crime via a cost-of-illness and jury compensation approach across several major crime categories. For this approach, we utilize the shooting victims estimations, which measure the reduction in the number of shooting victims as opposed to the reduction of shooting incidents. Our estimates find a reduction of 0.373 shooting victims per treated census block group, against a pre-treatment mean of 0.902, yielding an avoided 0.337 shooting victims per block group per year. Across the 143 treated census block groups in our sample, this implies approximately 48.2 shooting victims avoided annually due to the policy. While McCollister et al. (2010) does not provide a direct cost estimate for gunshot victimization, we use the societal costs of murder and aggravated assault as bounds. Drawing on the shooting victim dataset, in which approximately 19.4 percent of

shooting victims were fatal over the course of the sample, and assuming the policy-induced reduction maintained this same fatality rate, the avoided shooting victims decompose into approximately 9.4 murders and 38.8 nonfatal assault victims per year. Applying societal costs of \$8,982,907 per murder and \$107,020 per aggravated assault yields a total avoided societal cost of approximately \$88.6 million per year from shooting reductions alone. While our results also indicate significant reductions in the number gun crime incidents and Part 2 crime incidents, the heterogeneity of offense types within these categories, the variation in societal costs across individual offense types, and the inability to identify policy-induced responses at the offense level prevent a cost-of-crime estimate for these additional crime reductions.

A complementary approach to estimating the benefits of Focused Deterrence draws on our rental and housing market estimates through a revealed preference framework. To the extent that rental and housing markets reflect the desirability of a neighborhood, the capitalization of crime reductions into rental and housing prices reflects the marginal willingness to pay of residents and property owners for the bundle of amenity improvements produced by the policy, in this case the reduction of shootings, and other less serious crime offenses. Our estimates find that rental prices increased by approximately 6.0 to 7.6 percent and housing values increased by approximately 3.6 to 7.0 percent in treated census block groups relative to control block groups following the policy. Using the midpoints of these ranges and pre-treatment median values of \$799.50 for rent and \$229,732.20 for housing valuations, we can translate these estimates into aggregate dollar figures across the number of rental units and owner-occupied housing units in our treated group. These quantity of these rental units and owner-occupied housing units are estimated from the ACS 5-year data, and the number of units was estimated in 2012 (one-year prior to the implementation of the policy). For Southern Philadelphia, the 2012 ACS 5-year estimates suggest that there are around 30,255 renter-occupied units and 38,940 owner-occupied units. The estimated rental price increase implies approximately \$18.9 million in additional annual rental market value across treated communities, while the estimated housing value increase implies approximately \$474.1 million in aggregate housing wealth gains. As these figures reflect only the portion of policy benefits captured by local property markets, they should be interpreted as a conservative lower bound on the true

welfare gains from the policy.

9 Conclusion

This study investigates the impact of Philadelphia’s Focused Deterrence policy on crime reduction and its downstream effects on communities as measured through rental markets and housing valuations. We find significant declines in the number of shooting victims and gun crime incidents in treated communities, along with declines in overall crime incidents driven by reductions in less serious non-violent crime incidents. We find this reduction in crime is capitalized into rental prices and housing valuations, as both increase in the years following the policy.

Several limitations of this study are worth noting. First, while our matching procedure selects control census block groups based on demographic characteristics and long-run pre-policy crime trends, the possibility of treated and control block groups differing along unobservable dimensions that could influence crime or housing market outcomes cannot be ruled out. Second, there is no clear consensus on the most appropriate matching or weighting procedure for constructing a valid control group in this setting, and our results may be sensitive to the choice of matching specification—a concern we partially address through our alternative PSM specifications and IPW robustness checks, though no single approach can be declared definitively superior. Third, our downstream outcome measures are drawn from ACS 5-year estimates, which aggregate responses over a rolling five-year window and are therefore likely to attenuate the estimated effects of the policy, smoothing over the true dynamics of rental and housing market adjustment. A higher-frequency dataset with more precise geographic coverage would be needed to better capture the full magnitude and timing of these downstream effects.

While we measure downstream effects primarily through the rental and housing market, the reduction in crime can have other downstream effects such as commercial property capitalization, business starts and overall economic activity, or other avenues such as educational outcomes. The Philadelphia Focused Deterrence policy along with other place-based crime reduction policies are suitable contexts to investigate these downstream effects and remain important subjects for future

research.

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A Tables and Figures

A.1 Summary Statistics

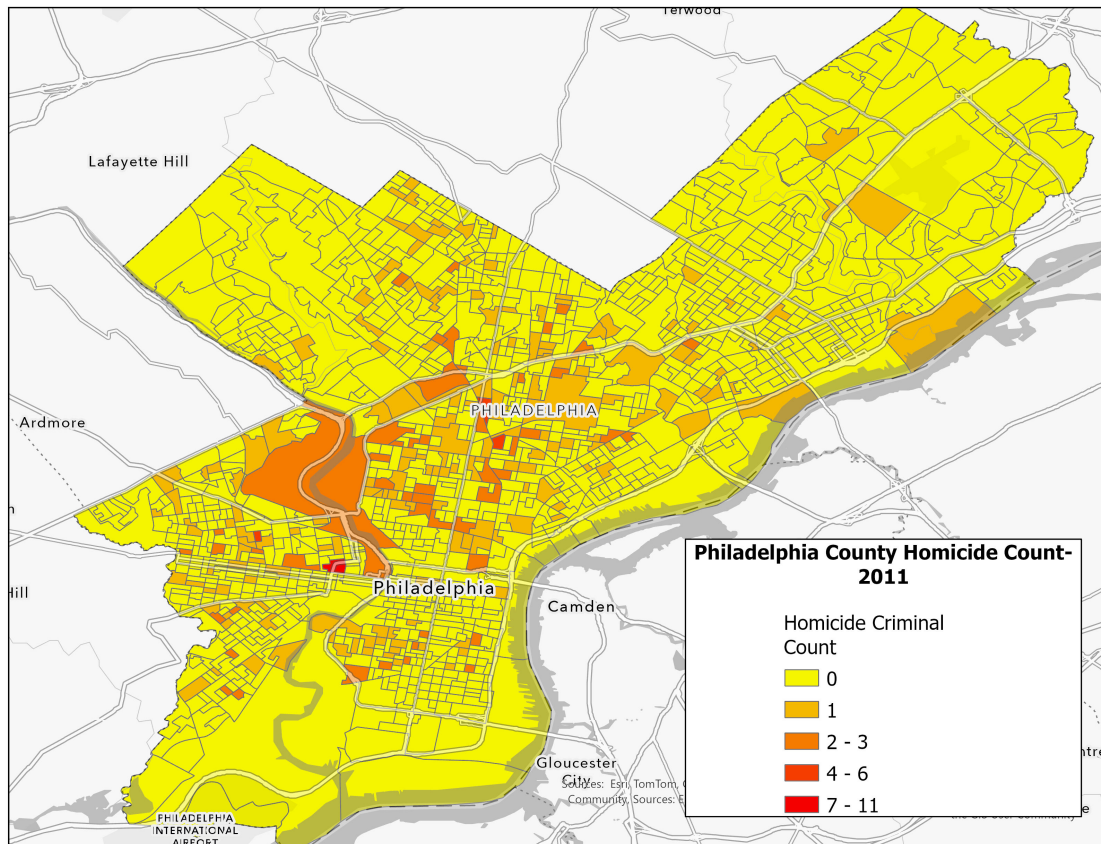


Figure 1: Philadelphia Homicide Crime Incidents, 2011

Table 1: Pre-Treatment (2010-2012) Crime and Community Characteristics Means

	(0) South Philadelphia	(1) Rest of Philadelphia	Difference (0) – (1)
<i>Panel A: Shootings and Crime Incidents</i>			
Shooting Victims	0.929 (0.082)	1.090 (0.030)	–0.161* (0.087)
Total Crime Incidents	134.505 (4.399)	140.233 (2.271)	–5.729 (4.951)
Part I Crime Incidents	50.299 (1.766)	54.855 (1.074)	–4.556** (2.067)
Part II Crime Incidents	84.205 (3.066)	85.378 (1.434)	–1.173 (3.384)
Gun Crime	4.416 (0.200)	5.933 (0.099)	–1.517*** (0.223)
<i>Panel B: Community Characteristics</i>			
Population	1123.822 (23.930)	1139.123 (8.841)	–15.301 (25.511)
% Black	0.293 (0.016)	0.494 (0.006)	–0.201*** (0.017)
% Hispanic	0.060 (0.004)	0.121 (0.003)	–0.061*** (0.005)
% Bachelor's or above	0.284 (0.011)	0.203 (0.003)	0.081*** (0.012)
Labor Force Participation	0.631 (0.006)	0.588 (0.002)	0.043*** (0.007)
Unemployment Rate	0.067 (0.003)	0.086 (0.001)	–0.019*** (0.003)
Share of Households on SNAP	0.110 (0.006)	0.143 (0.003)	–0.033*** (0.007)
Share of Households below Poverty Line	0.182 (0.008)	0.215 (0.003)	–0.033*** (0.009)
Median Household Income	\$42,275.61 (1026.760)	\$38,486.80 (350.093)	+\$3,788.81*** (1085.025)
Observations	438	3,546	3,984

Standard errors in parentheses.

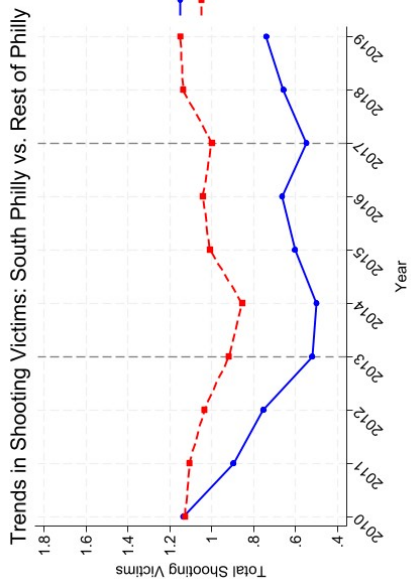
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 2: Pre-Treatment Means (2010-2012): Rental Prices and Housing Valuations

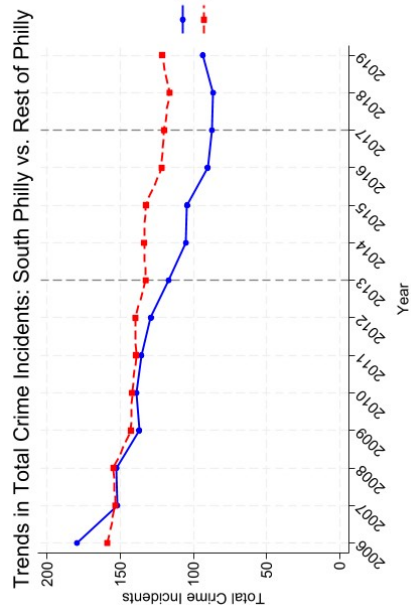
	(1) South Philadelphia	(0) Rest of Philadelphia	Difference (1) – (0)
Low Qrt. Rent	\$619.83 (11.165)	\$609.25 (4.334)	\$10.58 (11.977)
Median Rent	\$799.50 (12.768)	\$774.53 (4.761)	\$24.97* (13.626)
High Qrt. Rent	\$1,026.50 (17.824)	\$946.79 (5.965)	\$79.71*** (18.796)
Low Qrt. Housing Valuation	\$157,426.80 (5412.605)	\$120,664.50 (1687.135)	\$36,762.30*** (5669.455)
Median Housing Valuation	\$229,732.20 (7328.493)	\$165,754.00 (2284.047)	\$63,978.20*** (7676.176)
High Qrt. Housing Valuation	\$315,245.10 (10251.980)	\$221,073.50 (3098.045)	\$94,171.60*** (10709.854)
Observations	410	3,253	3,577

Standard errors in parentheses.

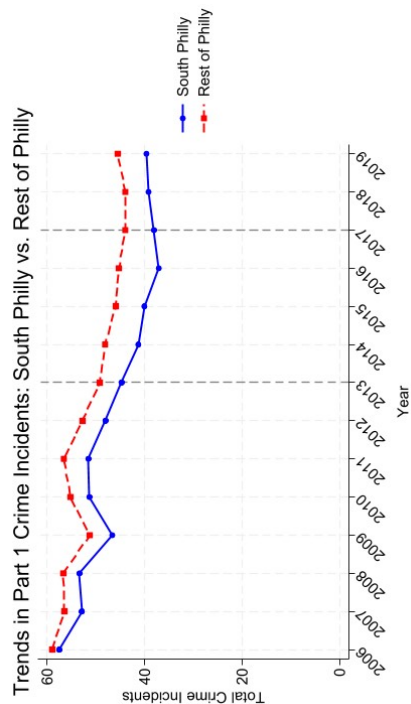
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$



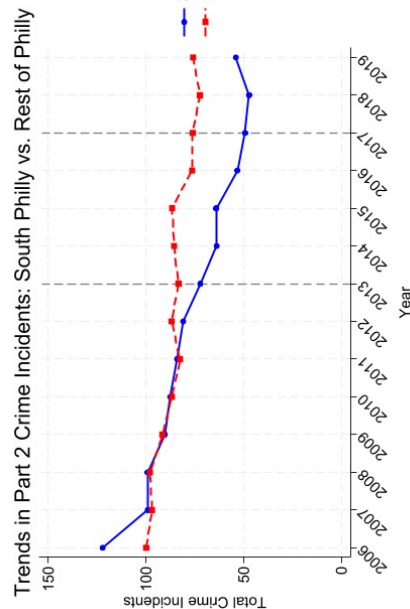
(a) Average Shooting Victims (2010–2019)



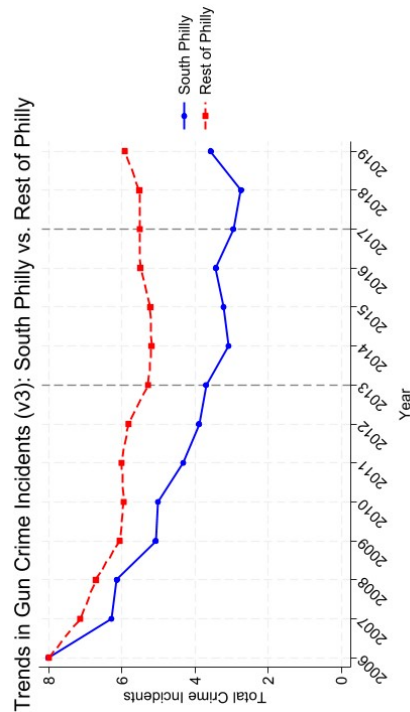
(b) Average Total Crime Incidents



(c) Average Part 1 Crime Incidents

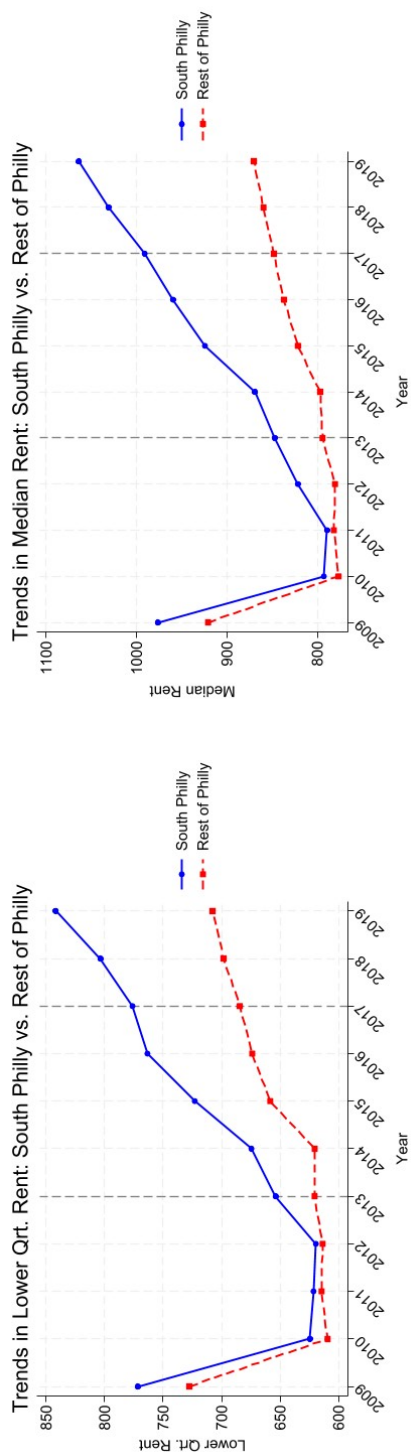


(d) Average Part 2 Crime Incidents



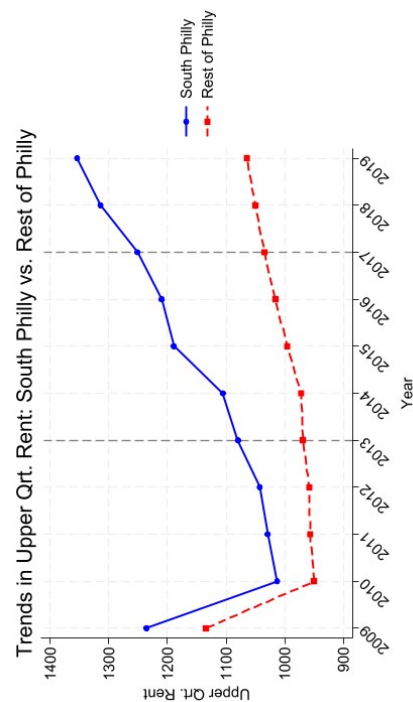
(e) Average Gun Crime (v3) Incidents

Figure 2: Pre-Matched Average Crime Incidents and Shooting Victims, 2006–2019



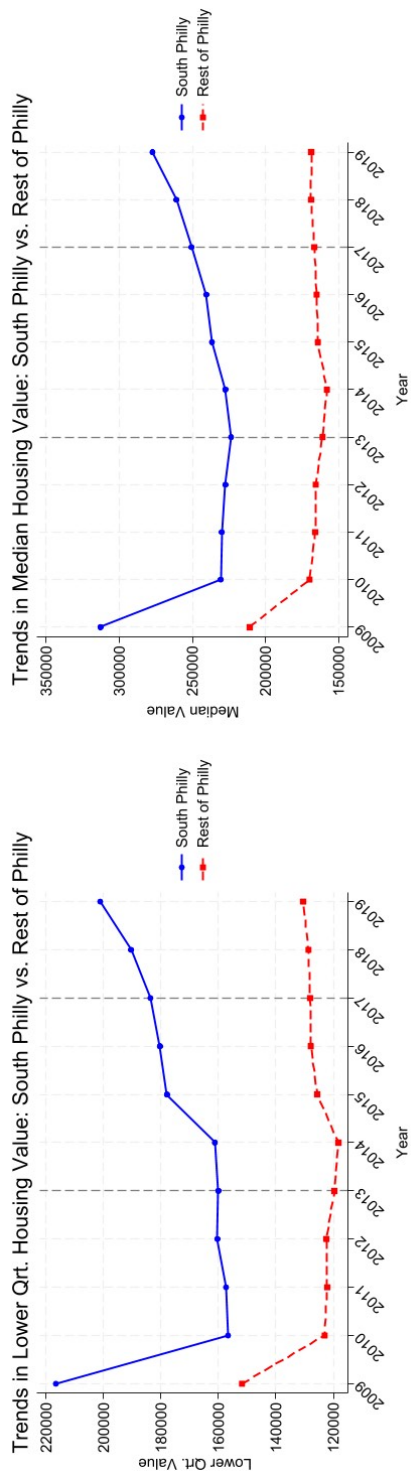
(a) Average Lower Qrt. Rental Prices

(b) Average Median Rental Prices



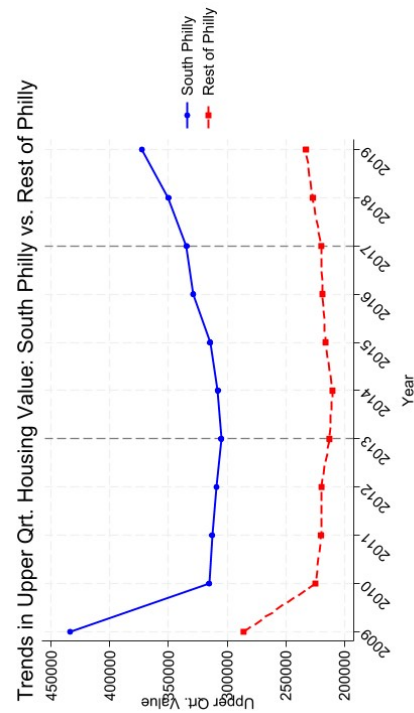
(c) Average High Qrt. Rental Prices

Figure 3: Pre-Matched Average Rental Prices



(a) Average Low Qrt. Housing Valuations

(b) Average Median Housing Valuations



(c) Average High Qrt. Housing Valuations

Figure 4: Pre-Matched Average Housing Valuations

A.2 PSM - Matching - Trends in Crime

Table 3: PSM Covariate Balance

Characteristic	Unmatched				Matched			
	Treated	Comp.	% Bias	<i>t</i> -stat.	Treated	Comp.	% Bias	<i>t</i> -stat.
<u>Crime Trends (2006–2012):</u>								
Part 1 Crime Count Trend	−9.466	−6.633	−14.4	−1.49	−8.867	−7.287	−8.0	−0.59
Part 2 Crime Count Trend	−41.212	−10.461	−41.4	−5.77	−33.462	−23.294	−13.7	−1.91
Gun Crime Count Trend	−4.123	−2.034	−37.4	−4.71	−3.881	−2.895	−17.7	−1.56
<u>Community Characteristics (Avg. 2009–2012):</u>								
Youth Male Density	0.065	0.077	−24.0	−2.34	0.065	0.070	−8.3	−0.73
Total Population	1133.5	1160.9	−5.3	−0.58	1136.6	1167.8	−6.1	−0.50
Percent Black	0.296	0.471	−49.2	−5.34	0.299	0.221	21.9	2.07
Percent Hispanic	0.060	0.124	−43.0	−3.88	0.060	0.052	5.4	0.88
Econ. Disadvantage Index	−0.228	−0.005	−29.9	−3.22	−0.219	−0.242	3.1	0.27
Number of Census Block Groups	146	1,182			143	119		

Notes: Balance statistics computed via `pstest` following nearest-neighbor propensity score matching. Matching procedure found the 1 nearest neighbor with a caliper of 0.03. % Bias is the standardized percentage bias in means between treated and comparison units. Rubin’s $B = 40.1$ for the matched sample. 3 treated block groups were dropped off common support.

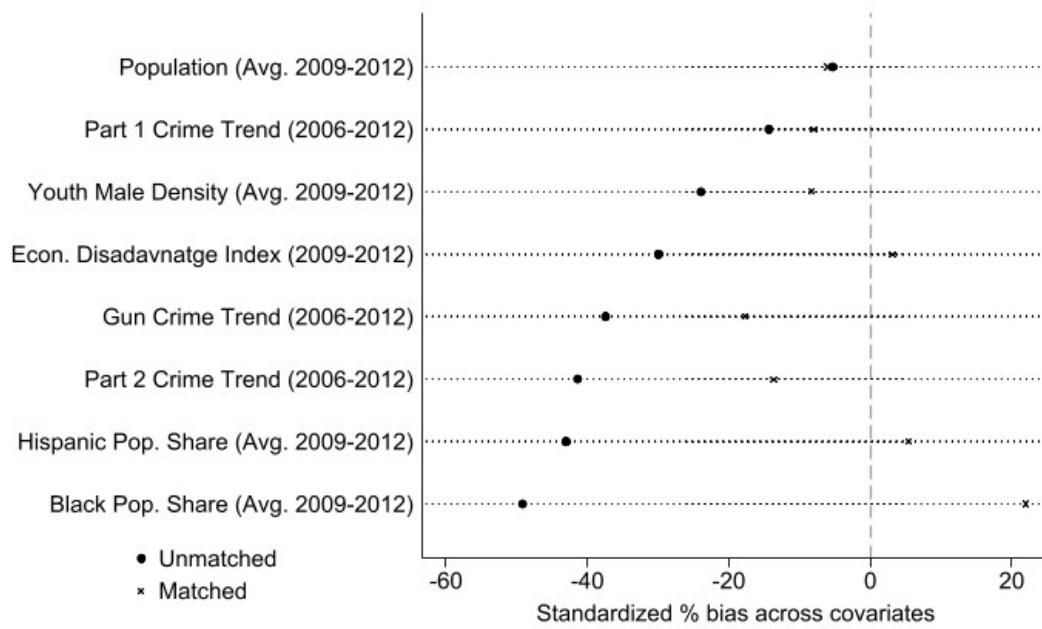


Figure 5: Covariate Balance Loveplot

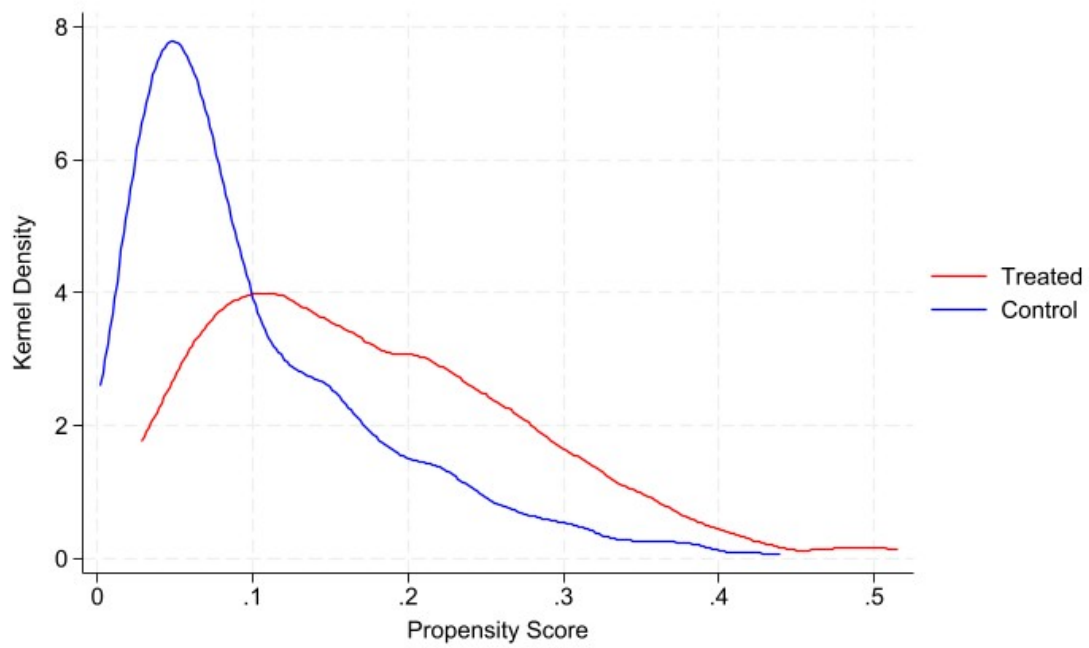


Figure 6: Kernel Density of Propensity Scores – Pre-Match

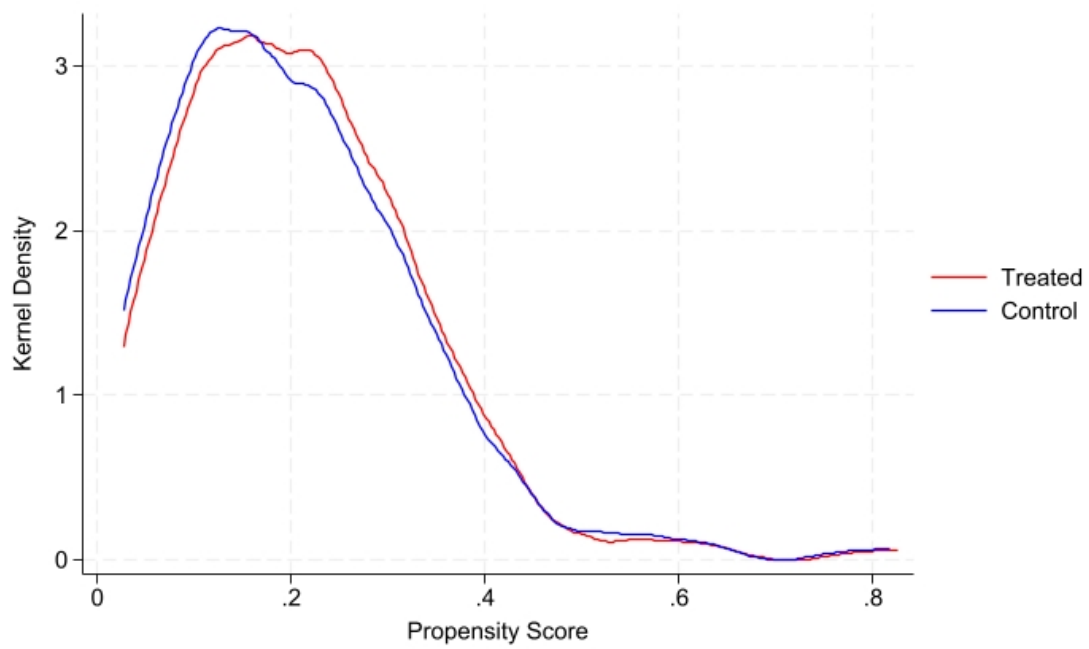


Figure 7: Kernel Density of Propensity Scores – Post-Match

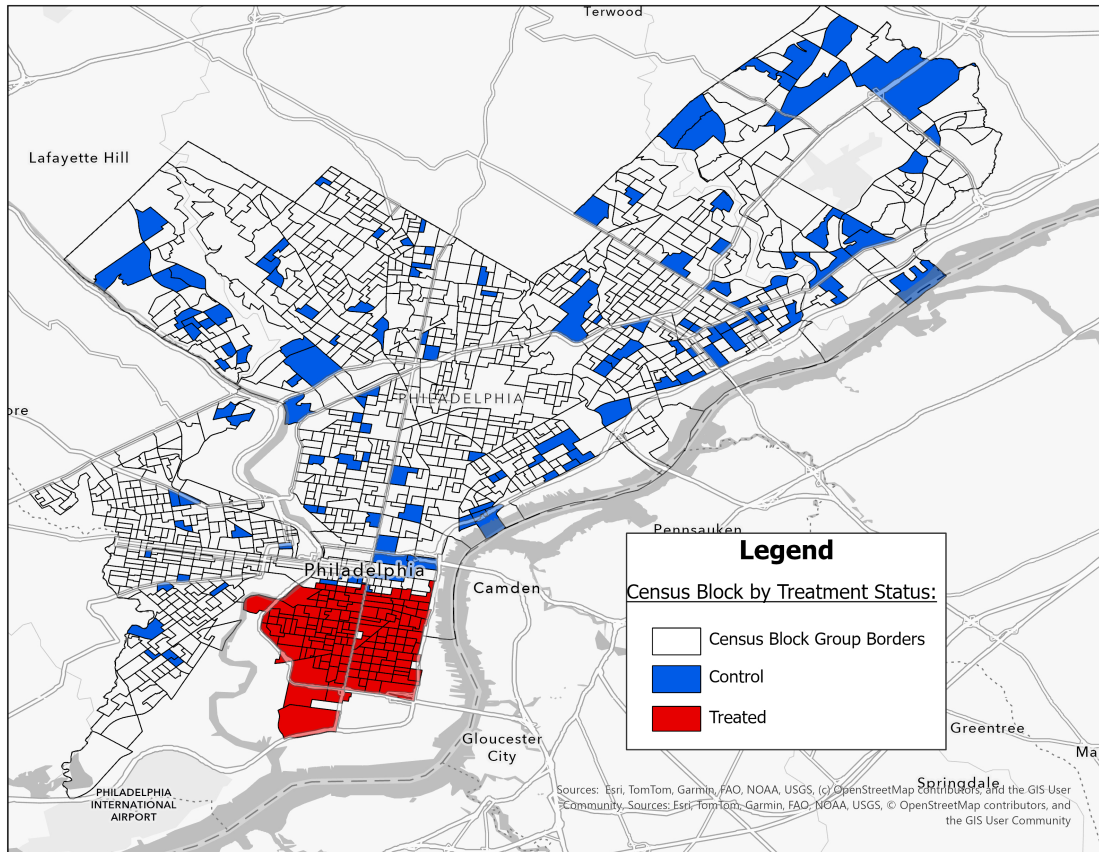
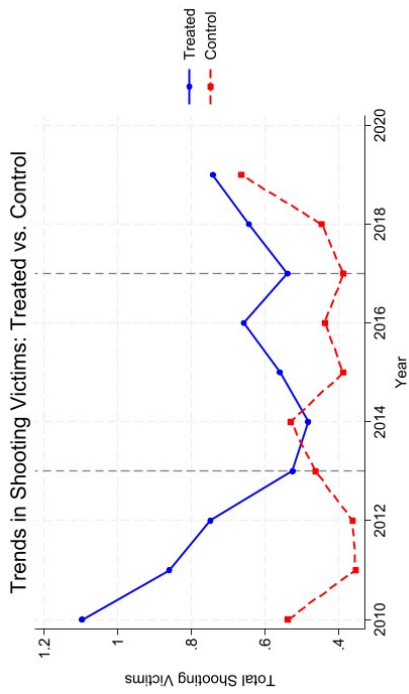
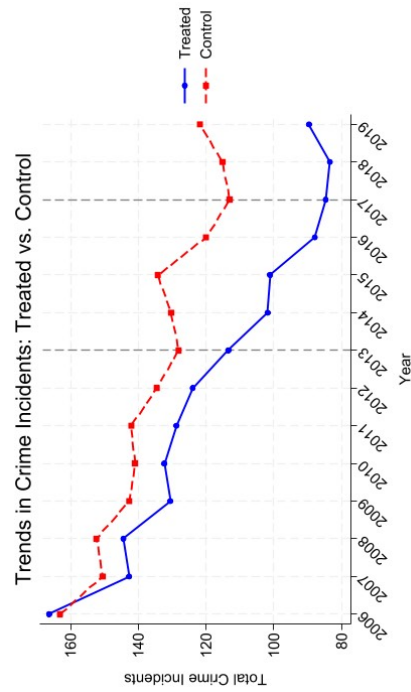


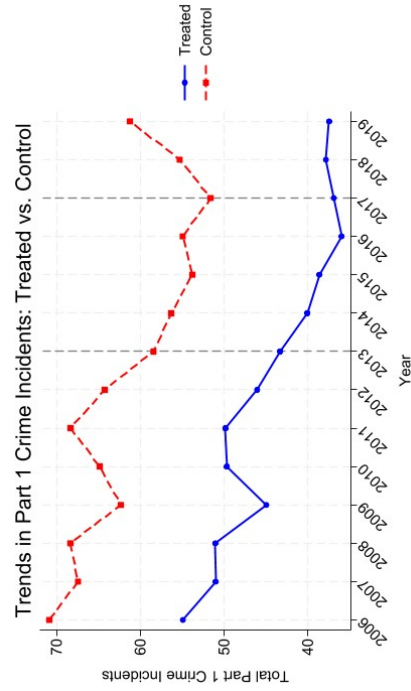
Figure 8: PSM Matched Treatment Map



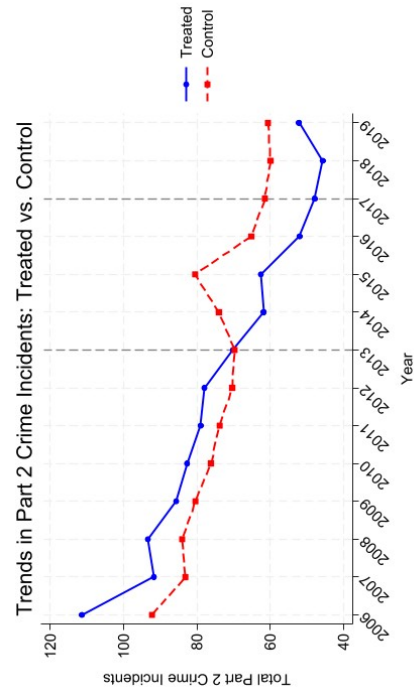
(a) Average Shooting Victims (2010–2019)



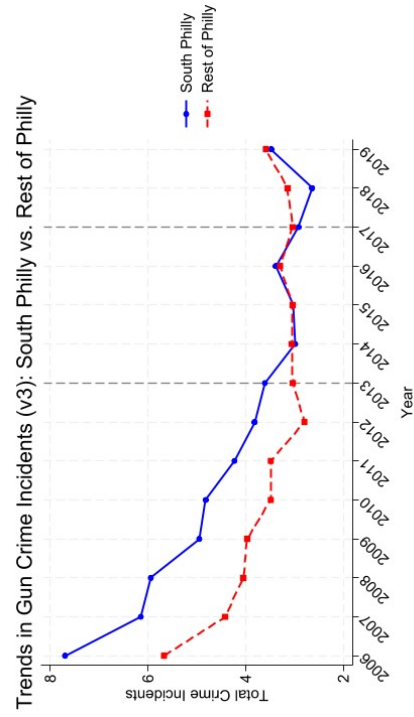
(b) Average Total Crime Incidents



(c) Average Part 1 Crime Incidents

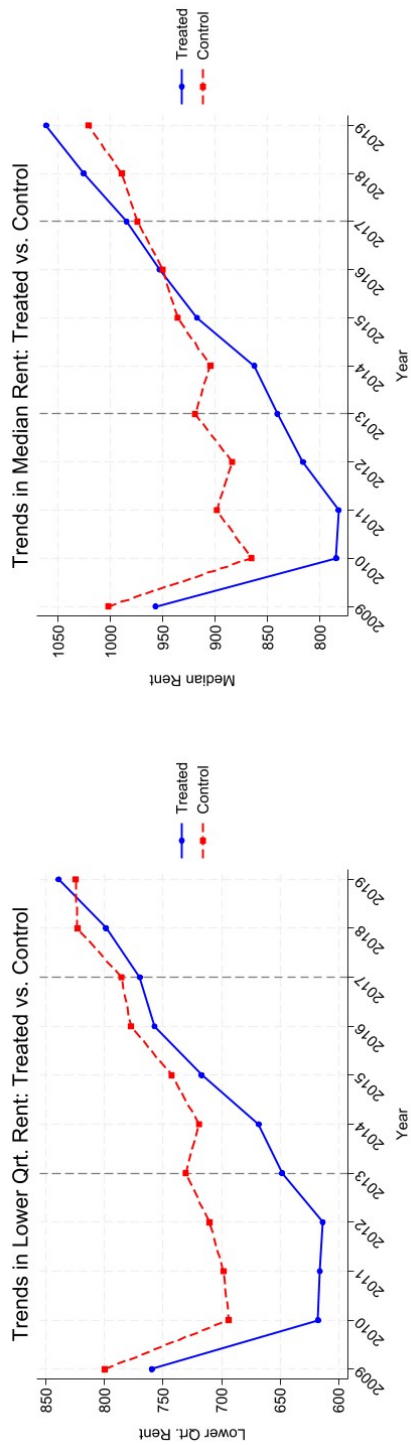


(d) Average Part 2 Crime Incidents



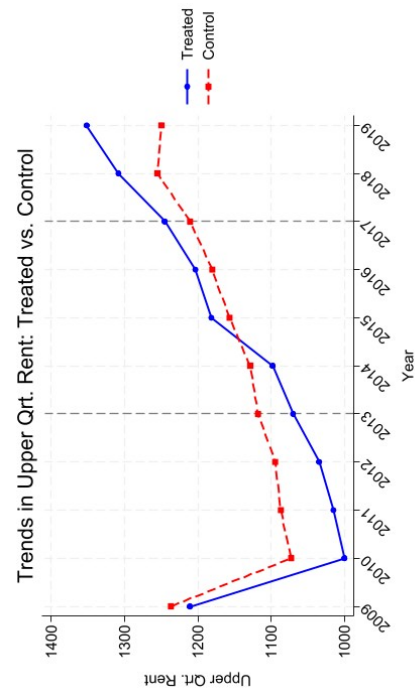
(e) Average Gun Crime (v3) Incidents

Figure 9: Matched Average Crime Incidents and Shooting Victims, 2006–2019



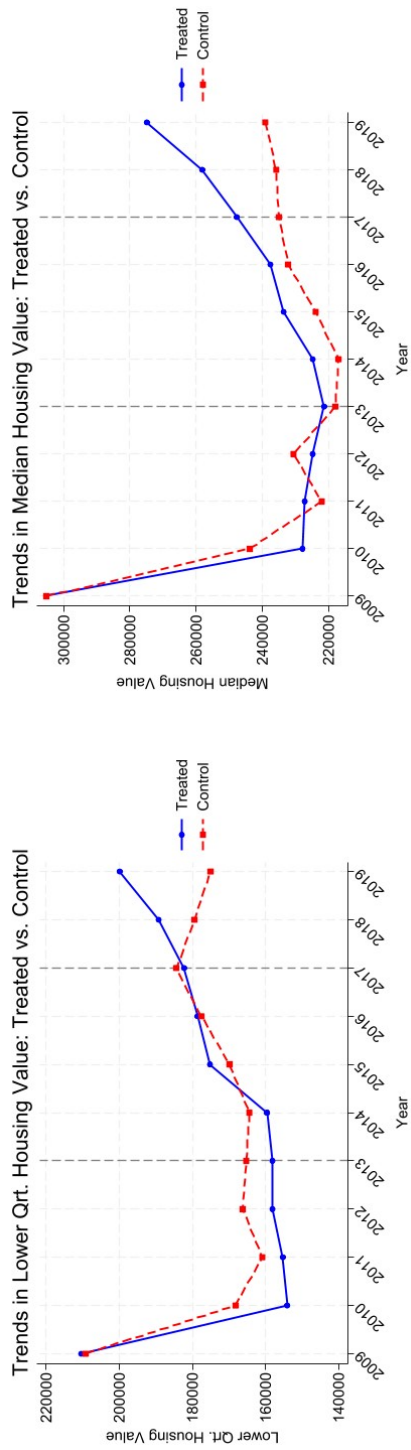
(b) Matched - Average Median Rental Prices

(a) Matched - Average Lower Qrt. Rental Prices



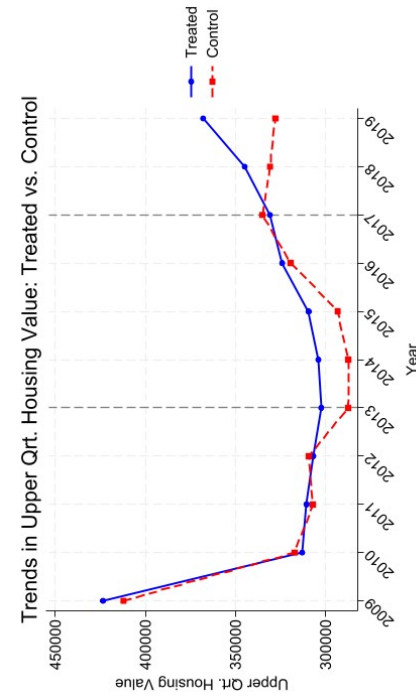
(c) Matched - Average High Qrt. Rental Prices

Figure 10: Matched Average Rental Prices



(b) Matched - Average Median Housing Valuations

(a) Matched - Average Low Qrt. Housing Valuations



(c) Matched - Average High Qrt. Housing Valuation

Figure 11: Matched Average Housing Valuations

Table 4: Crime Raw Means by Treatment Status and Period

	Control		Treated	
	Pre	Post	Pre	Post
Shooting Victims	0.179	0.473	0.387	0.592
Total Crime	146.61	123.23	138.43	94.58
Part 1 Crime	66.61	55.92	49.62	38.58
Part 2 Crime	79.99	67.31	88.81	56.00
Gun Crime	3.983	3.172	5.373	3.150

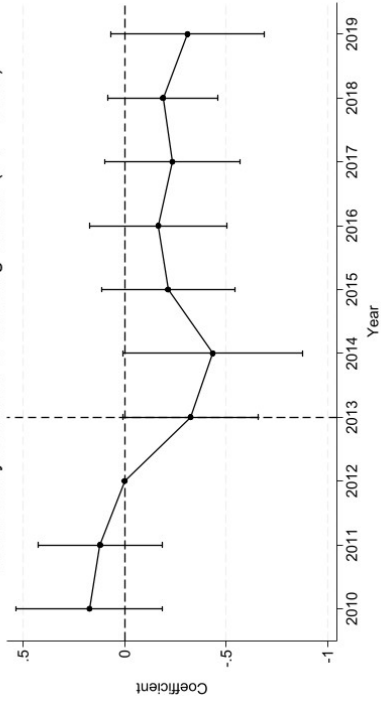
Notes: Raw means computed on the matched sample. Pre-treatment period is 2010–2012 for shooting victims, 2006–2012 for crime incidents; post-treatment period is 2013–2019. Treatment began in 2013.

Table 5: Raw Means by Treatment Status and Period: Rental Prices and Housing Valuations

	Control		Treated	
	Pre	Post	Pre	Post
Low Qrt. Rent	726.01	771.45	651.93	740.14
Median Rent	913.24	957.06	835.71	948.56
High Qrt. Rent	1,123.30	1,186.40	1,065.60	1,208.16
Low Qrt. Housing Val.	176,051	173,553	169,537	177,459
Median Housing Val.	250,280	228,649	246,398	242,500
High Qrt. Housing Val.	336,164	311,539	338,636	326,019

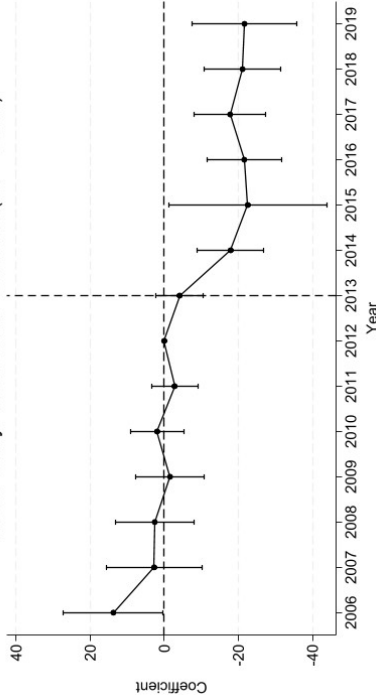
Notes: Raw means computed on the matched sample. Pre-treatment period is 2009–2012; post-treatment period is 2013–2019. Treatment began in 2013. Rents and housing valuations in 2019 dollars.

Event Study: Number of Shooting Victims (BY = 2012)



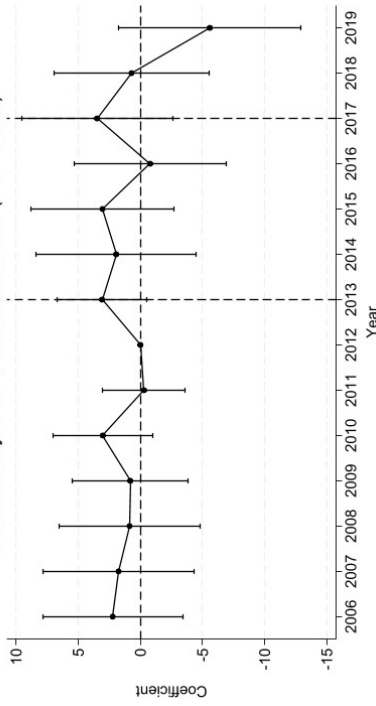
(a) Event Study - Shooting Victims

Event Study: Total Crime Incidents (BY = 2012)



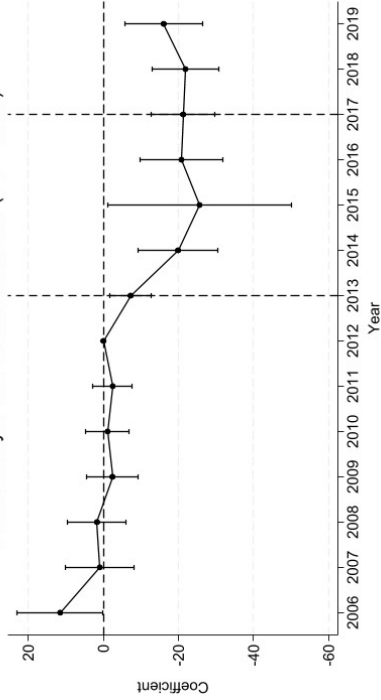
(b) Event Study - Total Crime Incidents

Event Study: Part 1 Crime Incidents (BY = 2012)



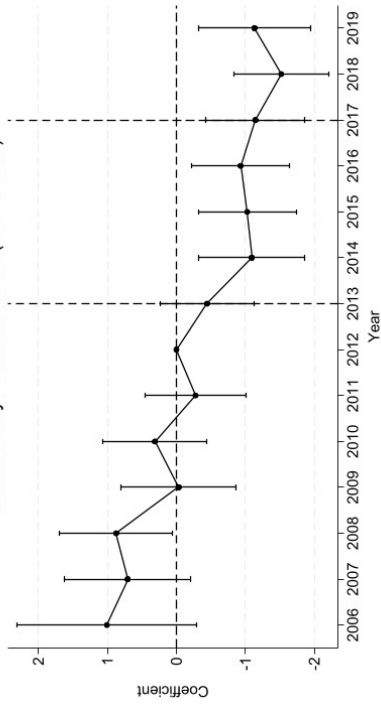
(c) Event Study - Part 1 Crime Incidents

Event Study: Part 2 Crime Incidents (BY = 2012)



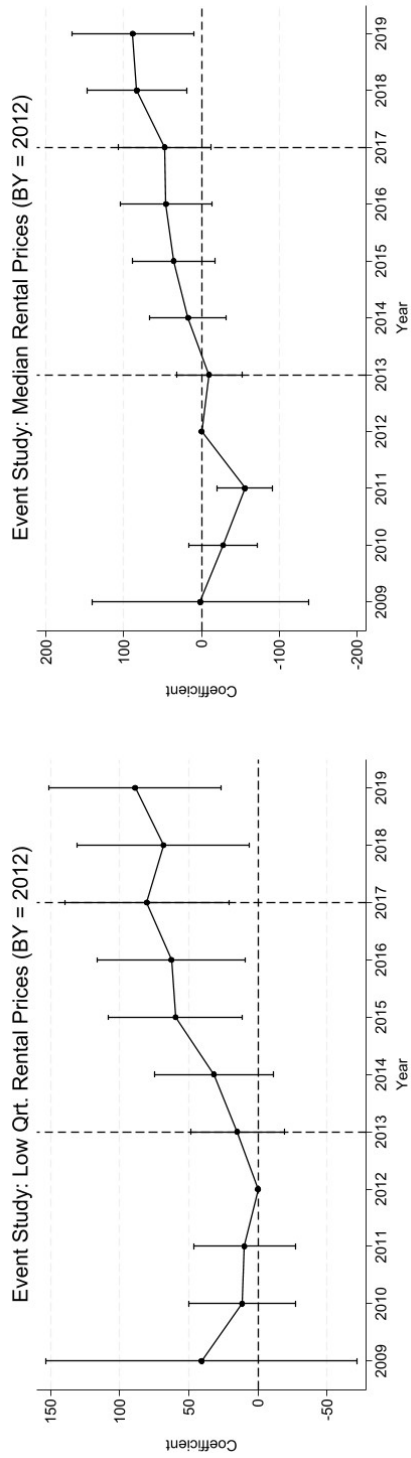
(d) Event Study - Part 2 Crime Incidents

Event Study: Gun Crime v3 (BY = 2012)



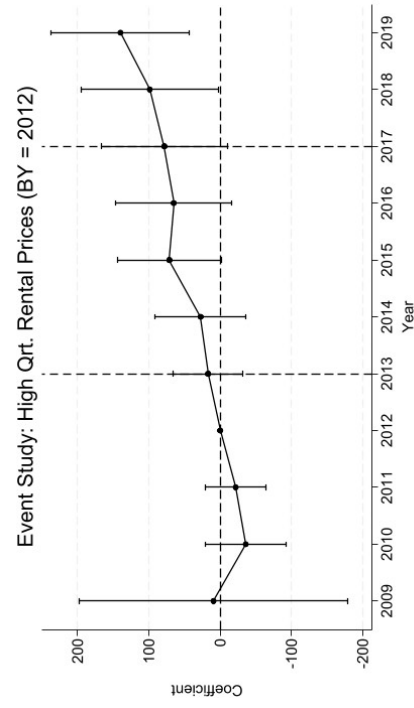
(e) Event Study - Gun Crime Incidents

Figure 12: Event Studies - Shooting Victims and Crime Incidents



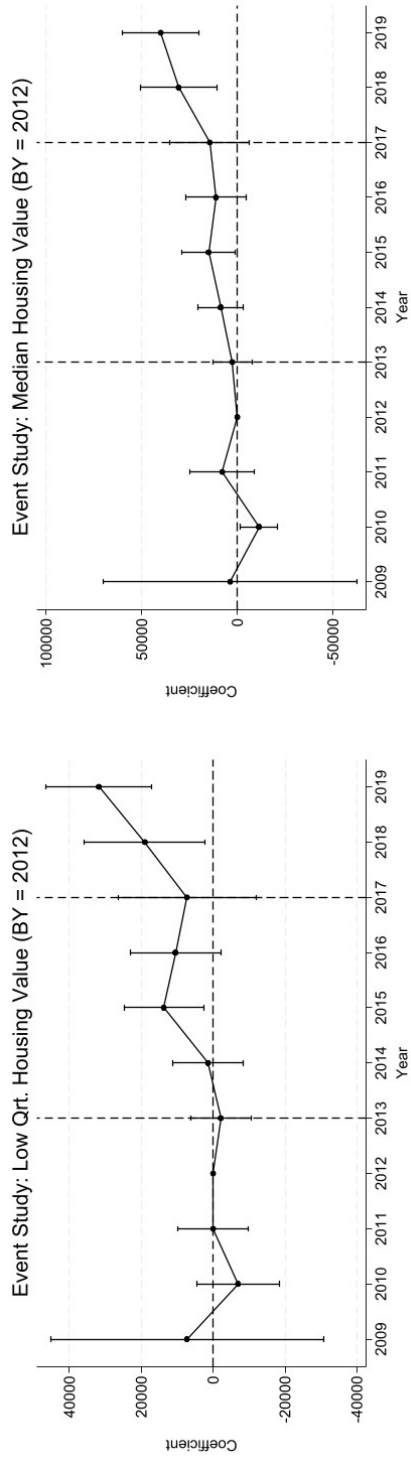
(a) Event Study - Low Qrt. Rent

(b) Event Study - Median Rent



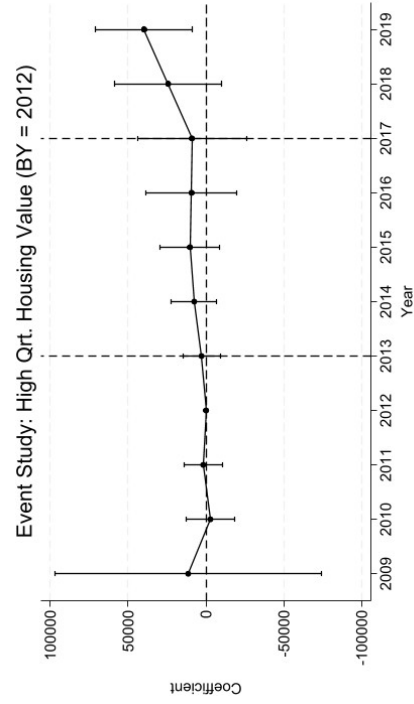
(c) Event Study - High Qrt. Rent

Figure 13: Event Studies - Rental Prices



(a) Event Study - Low Qrt. Housing Value

(b) Event Study - Median Housing Value



(c) Event Study - High Qrt. Housing Value

Figure 14: Event Study - Housing Value

Table 6: TWFE DiD: Effects of Focused Deterrence on Crime

	<i>Outcome Variable</i>				
	Shooting Victims	Total Crime	Part I Crime	Part II Crime	Gun Crime
Treat $\times$ Post	-0.373*** (0.091)	-19.165*** (4.489)	0.958 (2.679)	-20.123*** (4.762)	-1.298*** (0.247)
Census Block Group FEs	YES	YES	YES	YES	YES
Year FEs	YES	YES	YES	YES	YES
Pre-Treatment Mean	0.902	138.432	49.619	88.812	5.373
Sample	2010–2017	2006–2017	2006–2017	2006–2017	2006–2017
Observations	2,096	3,144	3,144	3,144	3,144
Within R-squared	0.0089	0.0182	0.0002	0.0237	0.0159

Notes: TWFE OLS estimates. Standard errors clustered at the block group level in parentheses. Sample restriction varies by outcome due to data availability. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 7: TWFE DiD Estimate: Part 1 Crime Categories

	(1) Part 1 Crimes	(2) Violent Crimes	(3) Property Crimes
Treat $\times$ Post	0.958 (2.675)	-1.859*** (0.488)	2.816 (2.350)
Census Block Group FEs	YES	YES	YES
Year FEs	YES	YES	YES
Pre-Treatment Mean	49.619	13.088	36.531
Sample	2006–2017	2006–2017	2006–2017
Observations	3,144	3,144	3,144
R^2	0.012	0.029	0.014

Notes: Robust standard errors clustered at the block group level in parentheses. All regressions include treated and post-period indicators. Pre-treatment means are calculated among treated block groups in the pre-period. Part 1 crimes include serious offenses against people and property. Violent crimes include homicide, sexual assault, robbery, and aggravated assault with and without a firearm. Property crimes include burglary, theft from a motor vehicle, larceny, auto theft, and arson. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 8: TWFE DiD Results: Part 2 Crimes

	(1) Part 2 Crimes	(2) Other Persons	(3) Public Disorder	(4) Drug & Alcohol	(5) White Collar	(6) Other
Treat $\times$ Post	-20.123*** (4.756)	-0.849 (0.580)	-3.867*** (1.224)	-2.933*** (0.594)	-1.597 (1.170)	-10.640*** (2.749)
Census Block Group FEs	YES	YES	YES	YES	YES	YES
Year FEs	YES	YES	YES	YES	YES	YES
Pre-Treatment Mean	88.812	17.074	20.119	9.185	6.626	32.393
Sample	2006–2017	2006–2017	2006–2017	2006–2017	2006–2017	2006–2017
Observations	3,144	3,144	3,144	3,144	3,144	3,144
R^2	0.018	0.004	0.030	0.082	0.014	0.020

Notes: Robust standard errors clustered at the block group level in parentheses. All regressions include treated and post-period indicators. Pre-treatment means are calculated among treated block groups in the pre-period. Part 2 crimes are less serious, generally nonviolent offenses. Other crimes against persons includes assaults with no resulting serious injury, sex offenses, and offenses against the family and children. Public disorder includes criminal mischief (vandalism), prostitution and commercialized vice, drunkenness, disorderly conduct, vagrancy, and curfew and loitering violations. Drug and alcohol offenses include drug abuse violations and liquor law offenses. White collar crimes include forgery, counterfeiting, fraud, and embezzlement. All other crimes include weapons offenses, stolen property offenses, and all other violations of state or local laws not otherwise specifically identified. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 9: TWFE DiD Estimates: Effects of Focused Deterrence on Rents and Housing Values

	<i>Outcome Variable</i>					
	Rent			Housing Value		
	Low Qrt.	Median	High Qrt.	Low Qrt.	Median	High Qrt.
Treat $\times$ Post	41.600* (23.153)	64.305** (27.565)	83.327** (37.336)	11,480 (6,582)	17,384* (10,488)	12,041 (16,206)
Census Block Group FEs	YES	YES	YES	YES	YES	YES
Year FEs	YES	YES	YES	YES	YES	YES
Pre-Treatment Mean	684.650	869.952	1,091.081	172,463	248,142	337,526
Sample	2009–2019	2009–2019	2009–2019	2009–2019	2009–2019	2009–2019
Observations	2,516	2,567	2,583	2,779	2,813	2,794
Within R-squared	0.0034	0.0055	0.0050	0.0024	0.0023	0.0006

Notes: TWFE OLS estimates. Standard errors clustered at the block group level in parentheses. Rents and housing valuations measured in dollars. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

B Robustness and Alt. Specifications

B.1 Comparison

Table 10: Robustness: Effects of Focused Deterrence on Crime Across Specifications

	<i>Outcome Variable</i>				
	Shooting Victims	Total Crime	Part I Crime	Part II Crime	Gun Crime
<i>Panel A: PSM – Trends (Main Specification)</i>					
Treat × Post	−0.373*** (0.091)	−19.165*** (4.489)	0.958 (2.679)	−20.123*** (4.762)	−1.298*** (0.247)
% Effect	−41.4%	−13.8%	1.9%	−22.7%	−24.2%
<i>Panel B: PSM – Levels Only</i>					
Treat × Post	−0.297*** (0.096)	−29.460*** (6.501)	−0.972 (1.502)	−28.488*** (6.462)	−1.349*** (0.250)
% Effect	−32.4%	−20.8%	−1.9%	−31.4%	−24.5%
<i>Panel C: PSM – Levels and Trends</i>					
Treat × Post	−0.253*** (0.094)	−9.690** (4.449)	−1.553 (1.211)	−8.137** (3.738)	−0.590* (0.320)
% Effect	−29.1%	−7.0%	−3.1%	−9.1%	−11.1%
<i>Panel D: AIPW Weighting</i>					
Treat × Post	−0.282*** (0.082)	−15.484*** (5.277)	1.250 (1.998)	−16.734*** (5.087)	−0.853*** (0.256)
% Effect	−30.4%	−10.6%	2.4%	−17.7%	−15.4%
Census Block Group FEs	YES	YES	YES	YES	YES
Year FEs	YES	YES	YES	YES	YES

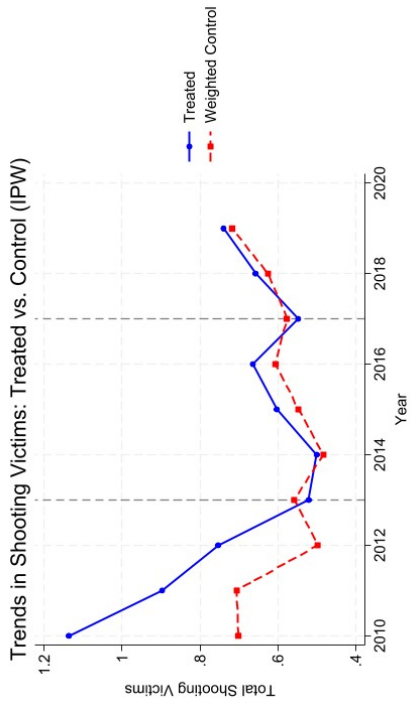
Notes: Each panel reports TWFE OLS estimates from a distinct comparison group construction. Panel A is the preferred specification, matching on block group demographics and pre-period crime trends (2006–2012). Panel B matches on demographics and average pre-period crime levels (2009–2012). Panel C matches on demographics, crime levels, and crime trends jointly. Panel D applies augmented inverse probability weights derived from the main propensity score model to the full Philadelphia sample. Percent effects computed as the coefficient divided by the panel-specific pre-treatment mean. Standard errors clustered at the block group level in parentheses. Sample periods: 2010–2017 (shooting victims) and 2006–2017 (all other outcomes). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 11: Robustness: Effects of Focused Deterrence on Rents and Housing Values Across Specifications

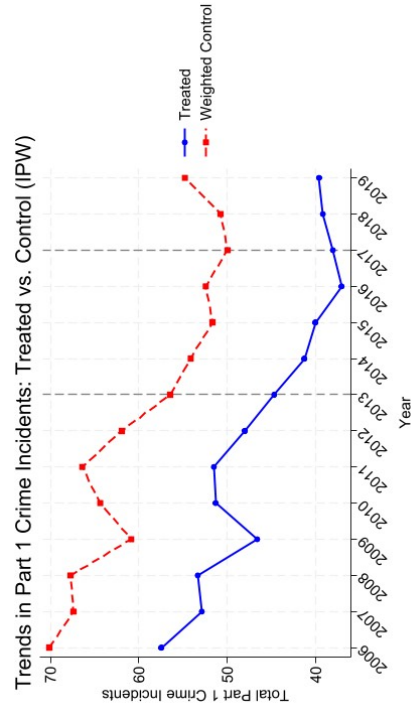
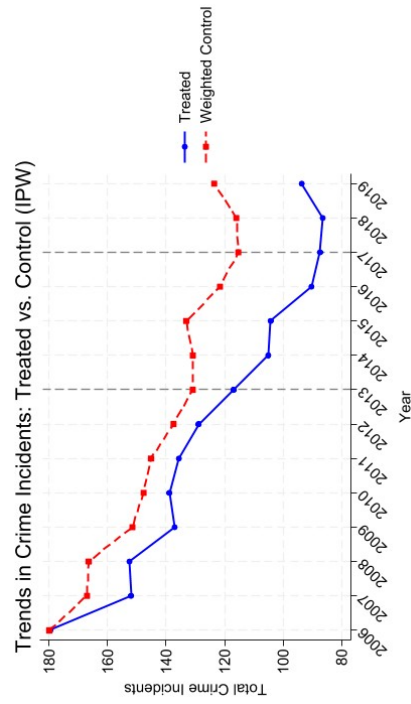
	<i>Outcome Variable</i>					
	Rent			Housing Value		
	Low Qrt.	Median	High Qrt.	Low Qrt.	Median	High Qrt.
<i>Panel A: PSM – Trends (Main Specification)</i>						
Treat × Post	41.600*	64.305**	83.327**	11,480	17,384*	12,041
	(23.153)	(27.565)	(37.336)	(6,582)	(10,488)	(16,206)
% Effect	6.1%	7.4%	7.6%	6.7%	7.0%	3.6%
<i>Panel B: PSM – Levels Only</i>						
Treat × Post	67.689***	106.411***	127.947***	20,725***	19,346**	16,005
	(23.202)	(26.389)	(35.641)	(6,627)	(9,431)	(14,110)
% Effect	9.8%	12.1%	11.7%	12.1%	8.0%	5.0%
<i>Panel C: PSM – Levels and Trends</i>						
Treat × Post	18.008	42.196	86.350**	7,433	8,769	2,467
	(23.479)	(25.674)	(35.742)	(6,379)	(8,549)	(14,797)
% Effect	2.7%	5.0%	8.1%	4.5%	3.7%	0.8%
<i>Panel D: AIPW Weighting</i>						
Treat × Post	43.753**	74.125***	109.359***	13,789**	14,175*	7,203
	(18.658)	(21.742)	(30.068)	(5,631)	(7,541)	(11,519)
% Effect	6.4%	8.4%	9.9%	7.9%	5.8%	2.2%
Census Block Group FEs	YES	YES	YES	YES	YES	YES
Year FEs	YES	YES	YES	YES	YES	YES
Sample	2009–2019					

Notes: Each panel reports TWFE OLS estimates from a distinct comparison group construction. Panel A is the preferred specification, matching on block group demographics and pre-period crime trends (2006–2012). Panel B matches on demographics and average pre-period crime levels (2009–2012). Panel C matches on demographics, crime levels, and crime trends jointly. Panel D applies augmented inverse probability weights derived from the main propensity score model to the full Philadelphia sample. Percent effects computed as the coefficient divided by the panel-specific pre-treatment mean. Standard errors clustered at the block group level in parentheses. Rents and housing valuations measured in dollars. *** p<0.01, ** p<0.05, * p<0.10.

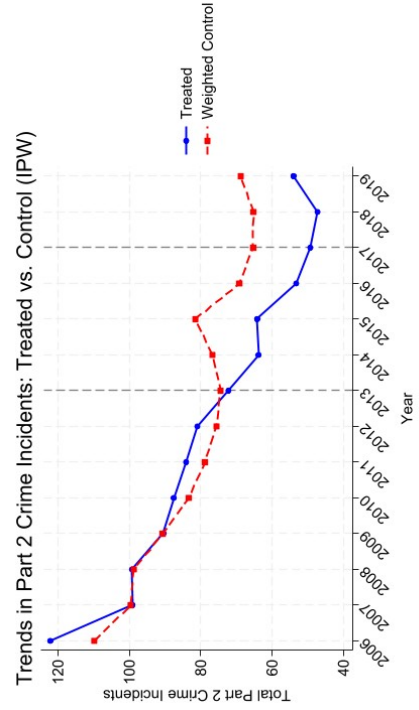
B.2 IPW - Trends



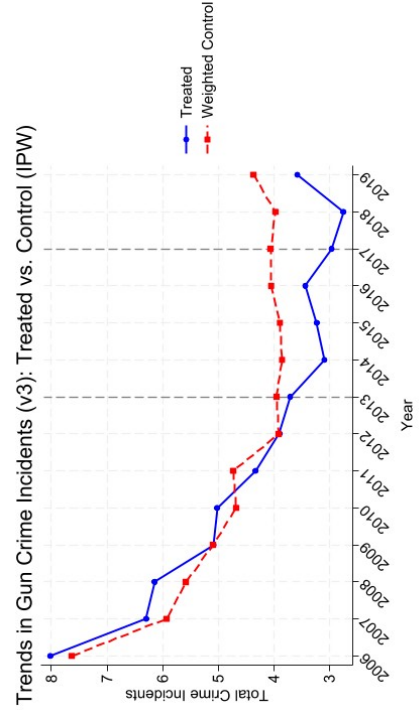
(a) Average Shooting Victims (2010–2019, Weighted)



(b) Average Total Crime Incidents, Weighted



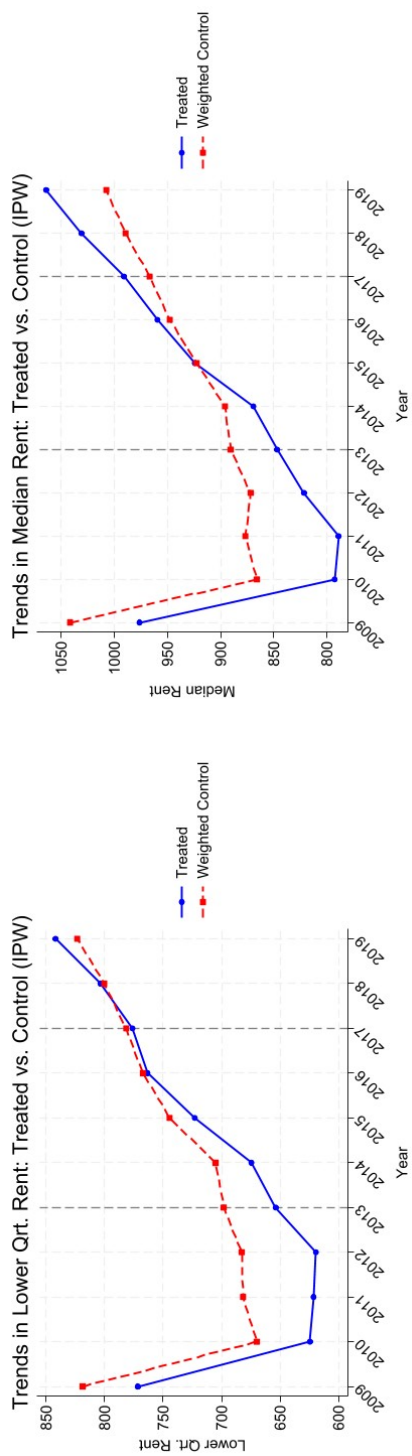
(c) Average Part 1 Crime Incidents, Weighted



(d) Average Part 2 Crime Incidents, Weighted

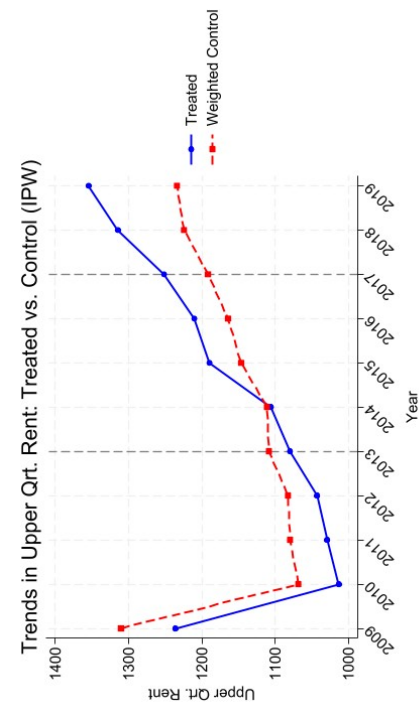
(e) Average Gun Crime (v3) Incidents, Weighted

Figure 15: IPW Weighted Average Crime Incidents and Shooting Victims, 2006–2019



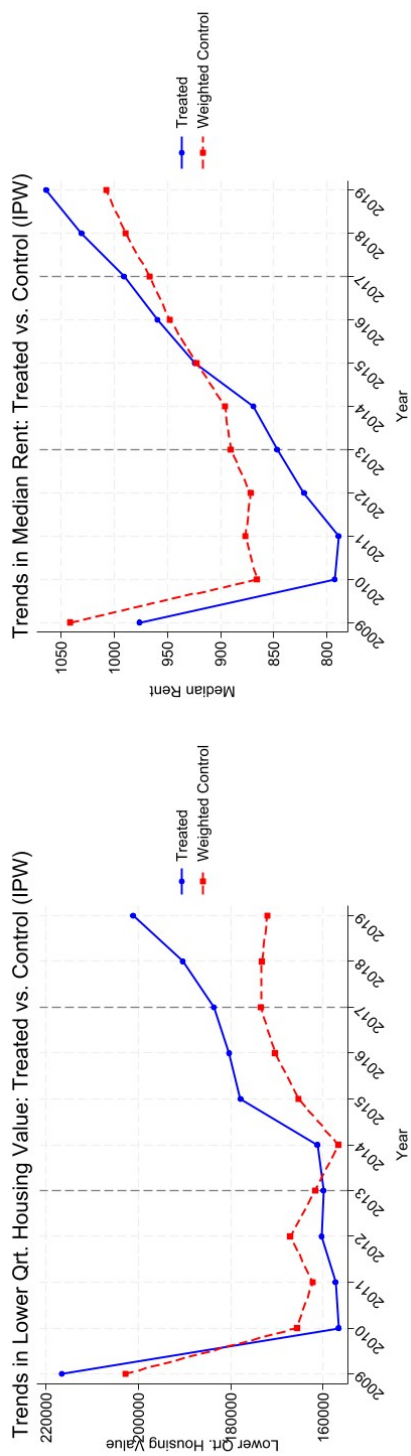
(a) Average Lower Qrt. Rental Prices, Weighted

(b) Average Median Rental Prices, Weighted



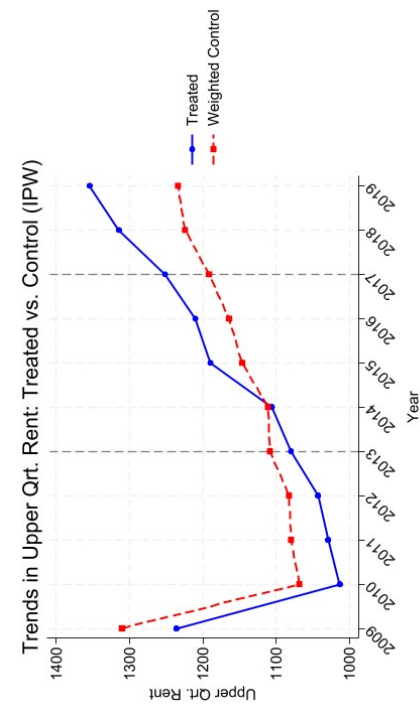
(c) Average High Qrt. Rental Prices, Weighted

Figure 16: IPW Weighted Average Rental Prices



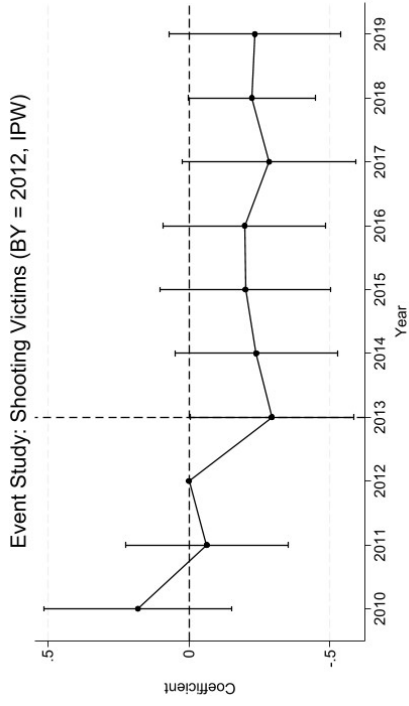
(a) Average Low Qrt. Housing Valuations, Weighted

(b) Average Median Housing Valuations, Weighted

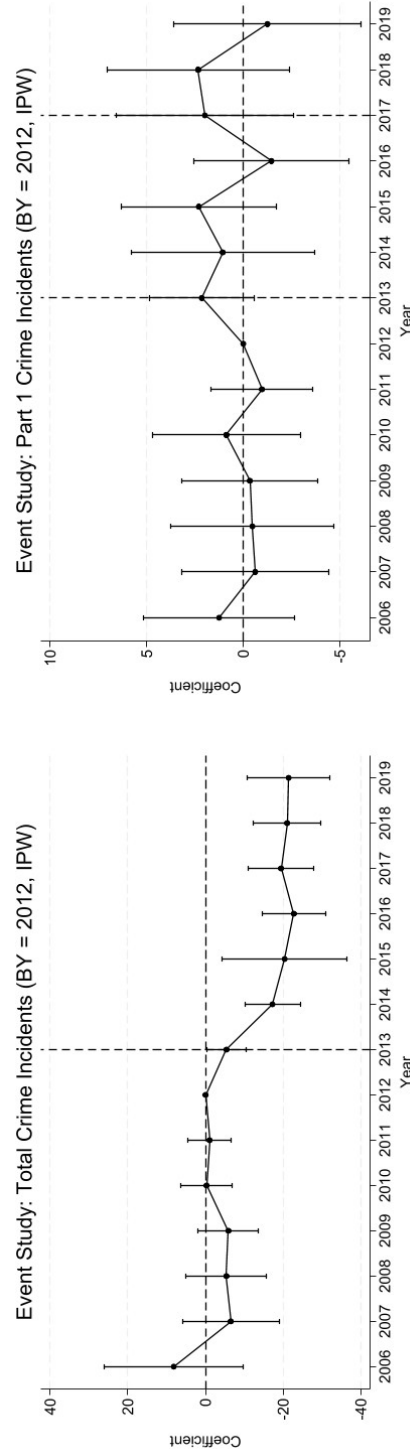


(c) Average High Qrt. Housing Valuations, Weighted

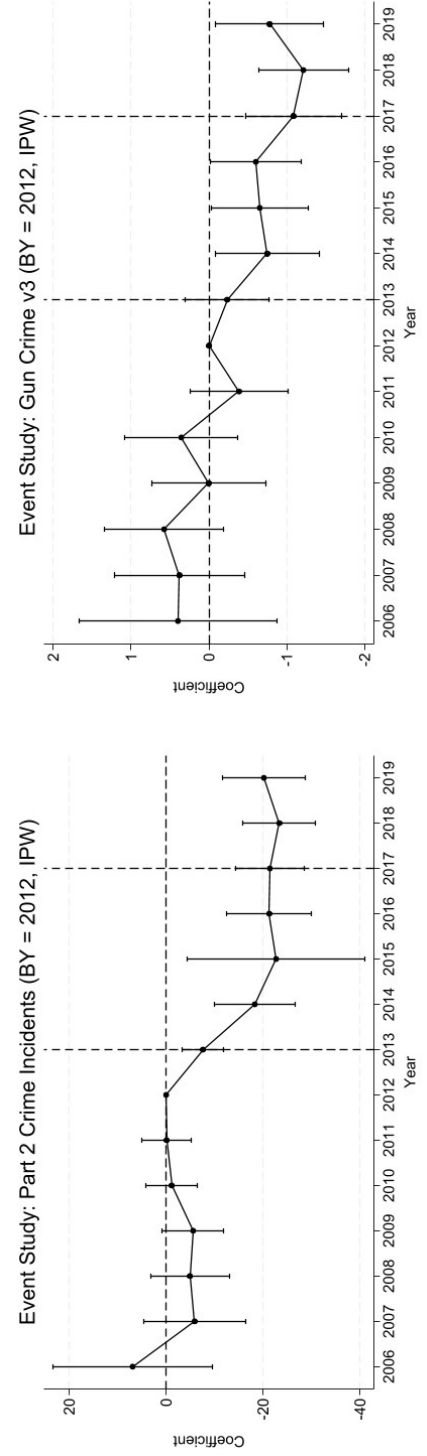
Figure 17: IPW Weighted Average Housing Valuations



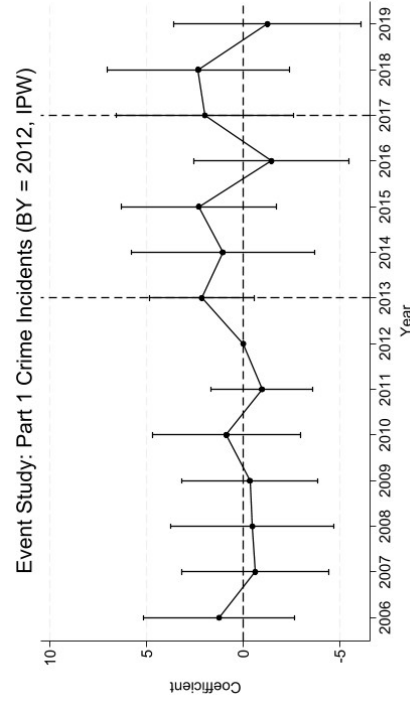
(a) IPW - Event Study - Shooting Victims



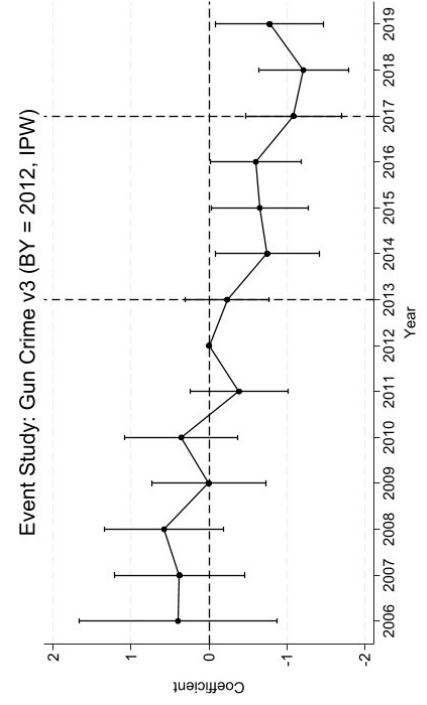
(b) IPW - Event Study - Total Crime Incidents



(d) IPW - Event Study - Part 2 Crime Incidents

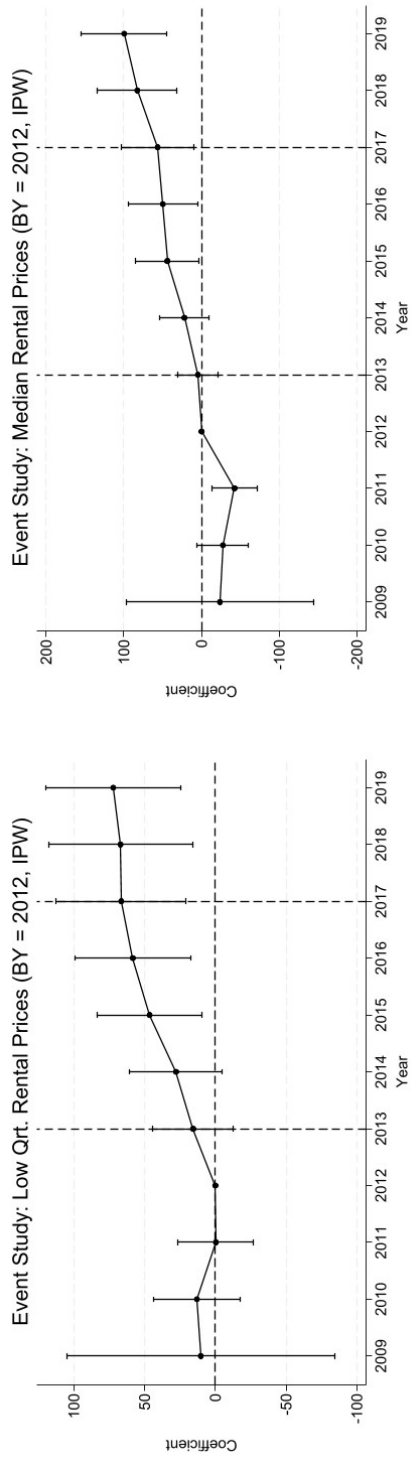


(c) IPW - Event Study - Part 1 Crime Incidents



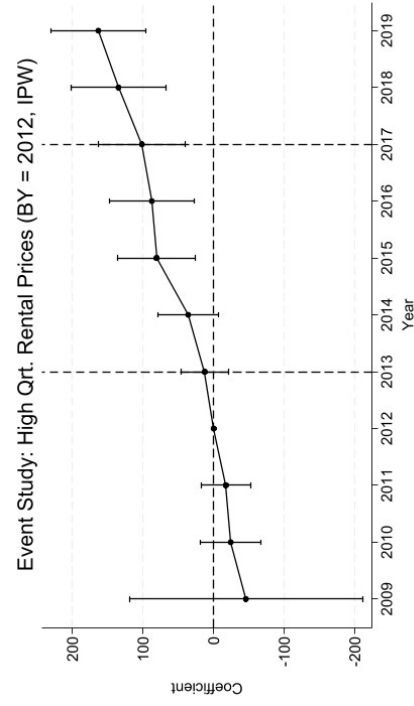
(e) IPW - Event Study - Gun Crime Incidents

Figure 18: IPW - Event Studies - Shooting Victims and Crime Incidents



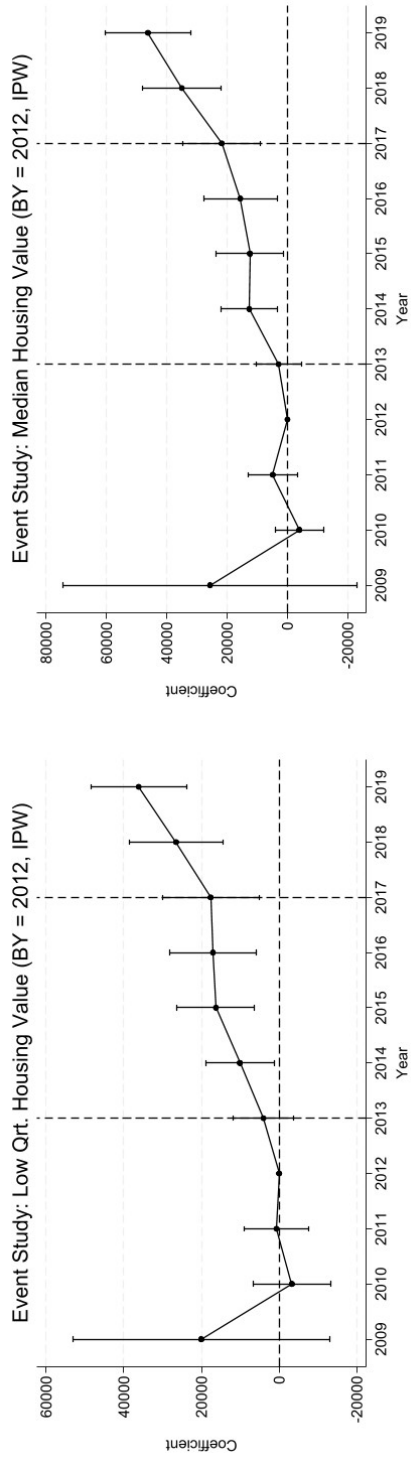
(a) IPW - Event Study - Low Qrt. Rent

(b) IPW - Event Study - Median Rent



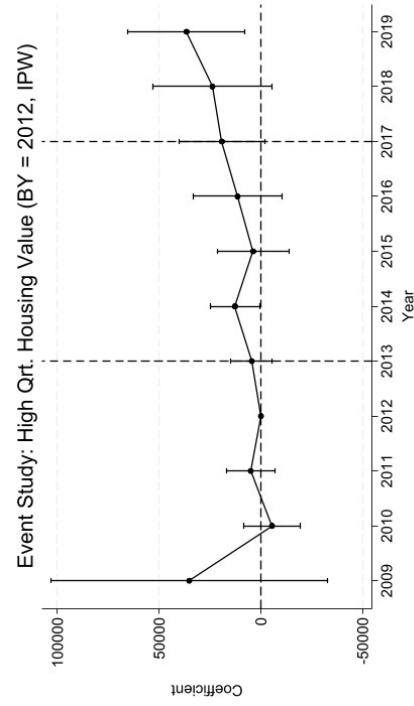
(c) IPW - Event Study - High Qrt. Rent

Figure 19: IPW - Event Studies - Rental Prices



(a) IPW - Event Study - Low Qrt. Housing Value

(b) IPW - Event Study - Median Housing Value



(c) IPW - Event Study - High Qrt. Housing Value

Figure 20: IPW - Event Study - Housing Value

Table 12: IPW-Weighted TWFE DiD Estimates: Effects of Focused Deterrence on Crime

	<i>Outcome Variable</i>				
	Shooting Victims	Total Crime	Part I Crime	Part II Crime	Gun Crime
Treat $\times$ Post	-0.282*** (0.082)	-15.484*** (5.277)	1.250 (1.998)	-16.734*** (5.087)	-0.853*** (0.256)
Census Block Group FEs	YES	YES	YES	YES	YES
Year FEs	YES	YES	YES	YES	YES
Pre-Treatment Mean	0.929	146.404	51.612	94.793	5.541
Sample	2010–2017	2006–2017	2006–2017	2006–2017	2006–2017
Observations	10,096	15,144	15,144	15,144	15,144
Within R-squared	0.0051	0.0085	0.0004	0.0119	0.0054

Notes: IPW-weighted TWFE OLS estimates using normalized ATT weights from a propensity score model. Standard errors clustered at the block group level in parentheses. Sample restriction varies by outcome due to data availability. *** p<0.01, ** p<0.05, * p<0.10.

Table 13: IPW-Weighted TWFE DiD Estimates: Effects of Focused Deterrence on Rents and Housing Values

	<i>Outcome Variable</i>					
	Rent			Housing Value		
	Low Qrt.	Median	High Qrt.	Low Qrt.	Median	High Qrt.
Treat $\times$ Post	43.753** (18.658)	74.125*** (21.742)	109.359*** (30.068)	13,789** (5,631)	14,175* (7,541)	7,203 (11,519)
Census Block Group FEs	YES	YES	YES	YES	YES	YES
Year FEs	YES	YES	YES	YES	YES	YES
Pre-Treatment Mean	685.948	879.102	1,107.525	173,631	244,647	330,127
Sample	2009–2019	2009–2019	2009–2019	2009–2019	2009–2019	2009–2019
Observations	11,836	12,222	12,325	13,083	13,292	13,157
Within R-squared	0.0034	0.0068	0.0084	0.0033	0.0018	0.0002

Notes: IPW-weighted TWFE OLS estimates using normalized ATT weights from a propensity score model. Standard errors clustered at the block group level in parentheses. Rents and housing valuations measured in dollars. *** p<0.01, ** p<0.05, * p<0.10.

B.3 Alternative PSM Matching

B.3.1 Levels

Characteristic	Unmatched				Matched			
	Treated	Comp.	% Bias	<i>t</i> -stat.	Treated	Comp.	% Bias	<i>t</i> -stat.
<u>Crime Outcomes (Avg. 2009–2012):</u>								
Part 1 Crime Count	49.384	52.709	−6.6	−0.64	48.517	55.247	−13.4	−1.15
Part 2 Crime Count	85.767	84.726	1.4	0.15	83.107	92.616	−12.9	−0.90
Gun Crime Count	4.582	5.683	−24.0	−2.49	4.547	4.443	2.3	0.20
<u>Community Characteristics (Avg. 2009–2012):</u>								
Youth Male Density	0.065	0.077	−24.0	−2.34	0.065	0.068	−4.2	−0.41
Total Population	1133.5	1160.9	−5.3	−0.58	1132.2	1151.3	−3.7	−0.33
Percent Black	0.296	0.471	−49.2	−5.34	0.298	0.272	7.3	0.65
Percent Hispanic	0.060	0.124	−43.0	−3.88	0.060	0.067	−5.0	−0.75
Econ. Disadvantage Index	−0.228	−0.005	−29.9	−3.22	−0.222	−0.244	3.0	0.27
Number of Census Block Groups	146	1,182			145	132		

Notes: Balance statistics computed via `pstest` following nearest-neighbor propensity score matching. Matching procedure found the 1 nearest neighbor with a caliper of 0.03. % Bias is the standardized percentage bias in means between treated and comparison units. Rubin’s $B = 23.0$ for the matched sample. 1 treated block groups were dropped off common support.

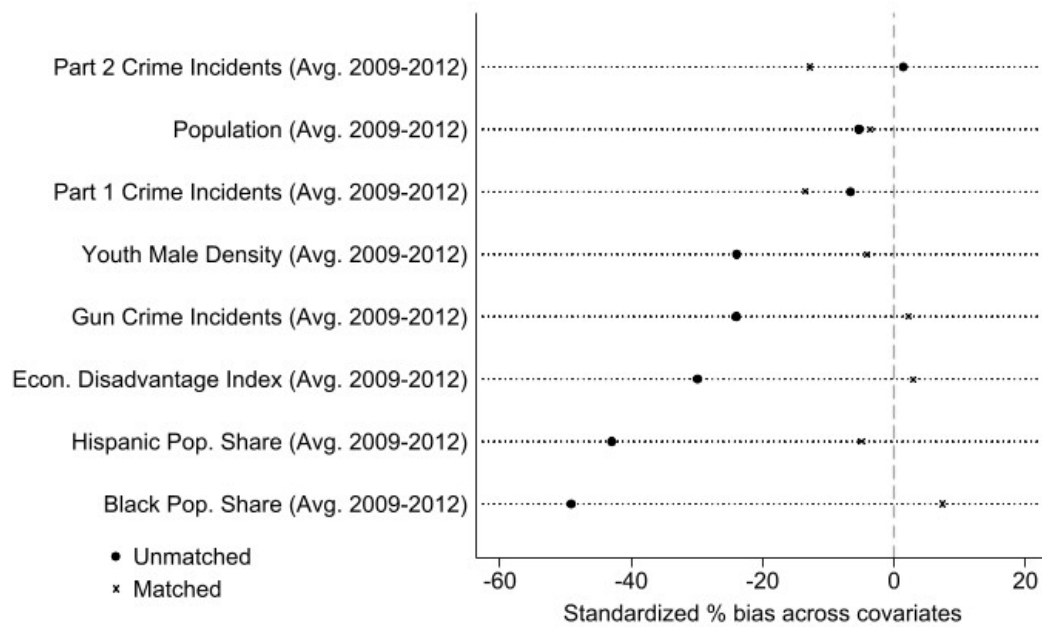


Figure 21: Covariate Balance Loveplot - Levels

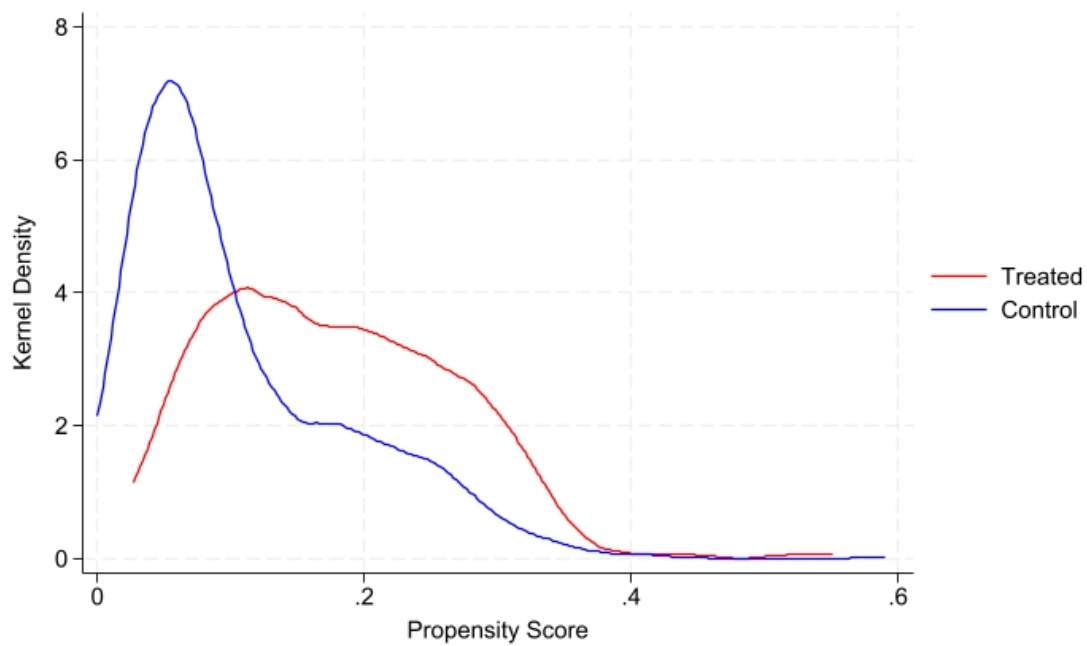


Figure 22: Kernel Density of Propensity Scores – Pre-Match - Levels

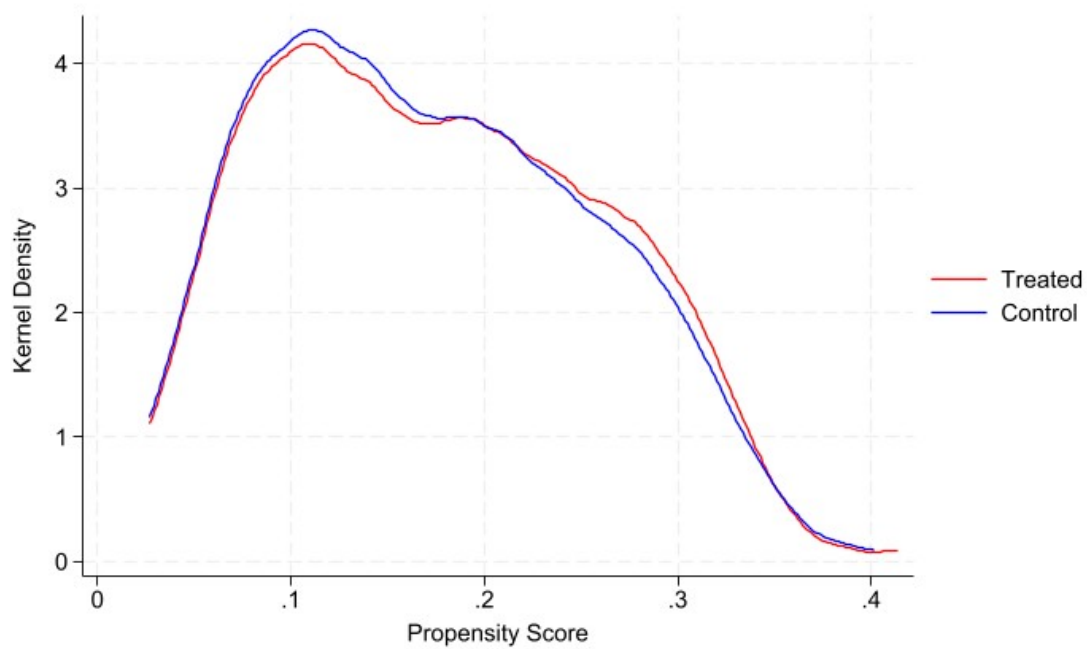
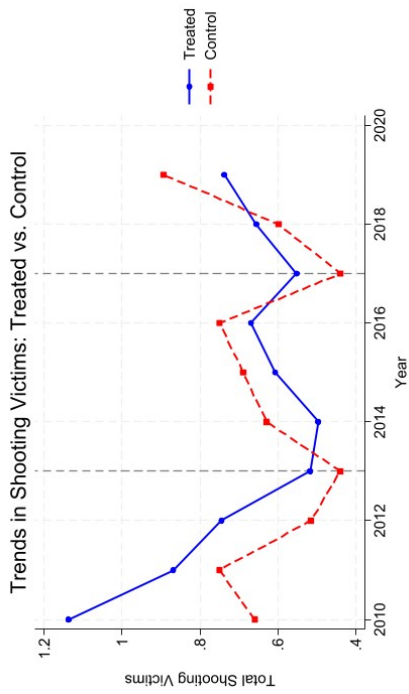
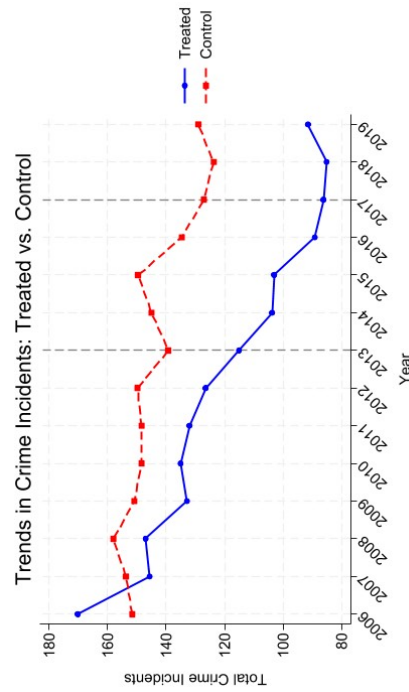


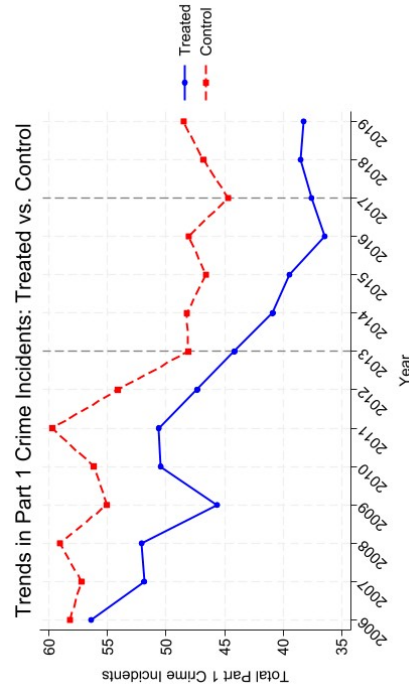
Figure 23: Kernel Density of Propensity Scores – Post-Match - Levels



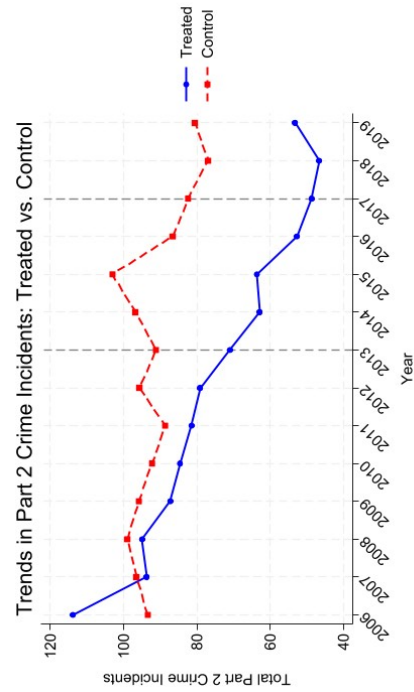
(a) Average Shooting Victims (2010–2019) - Levels



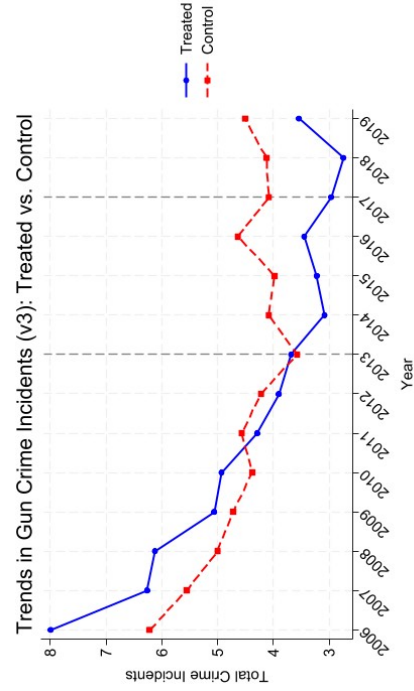
(b) Average Total Crime Incidents - Levels



(c) Average Part 1 Crime Incidents - Levels

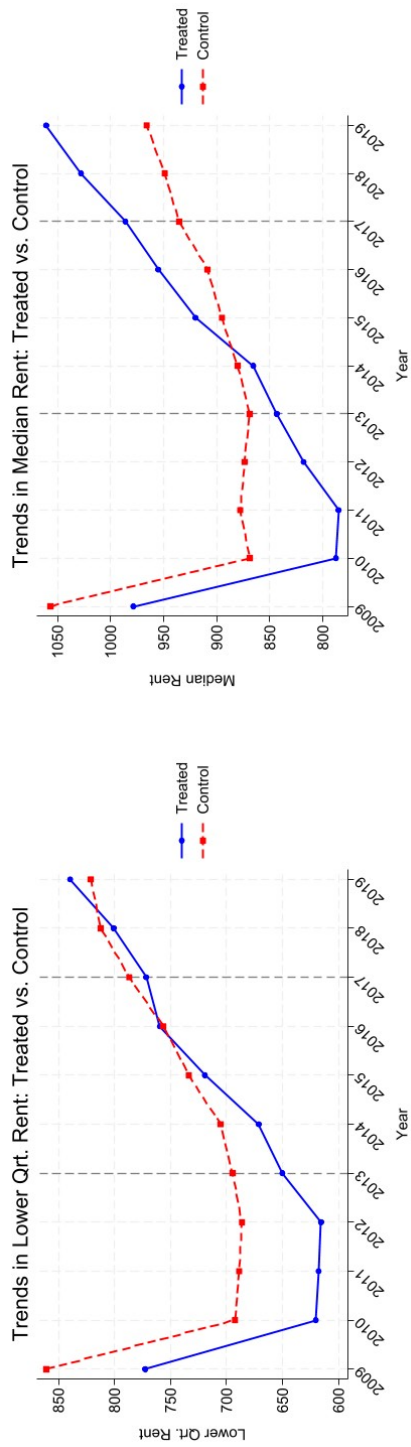


(d) Average Part 2 Crime Incidents - Levels



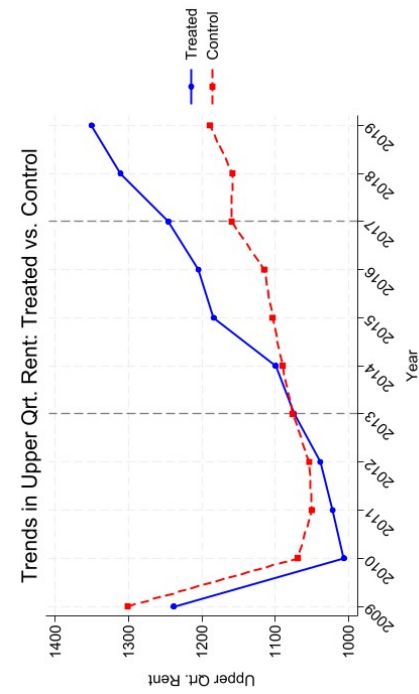
(e) Average Gun Crime (v3) Incidents - Levels

Figure 24: Post-Match Average Crime Incidents and Shooting Victims, 2006–2019 - Levels



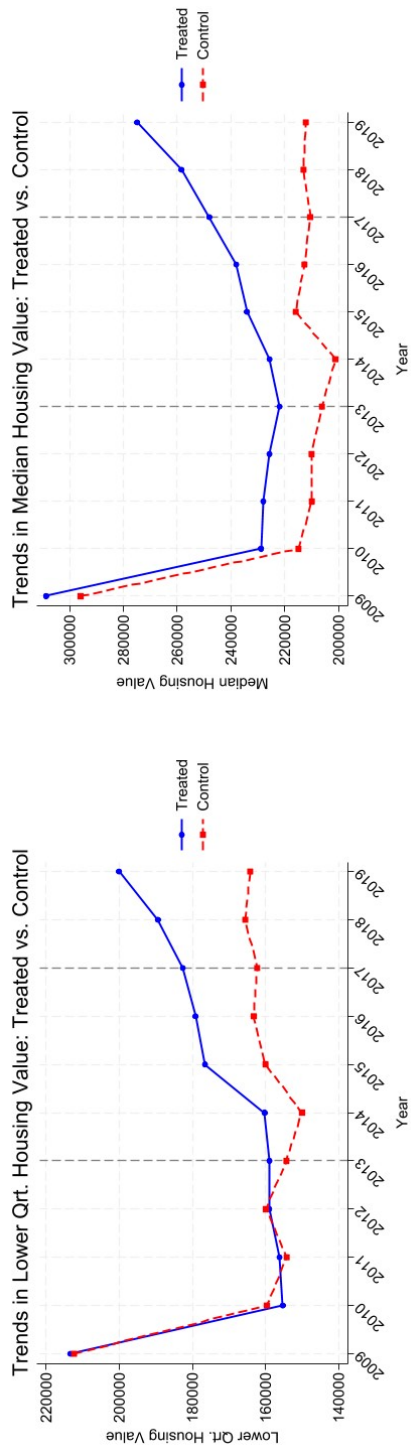
(a) Average Lower Qrt. Rental Prices - Levels

(b) Average Median Rental Prices - Levels



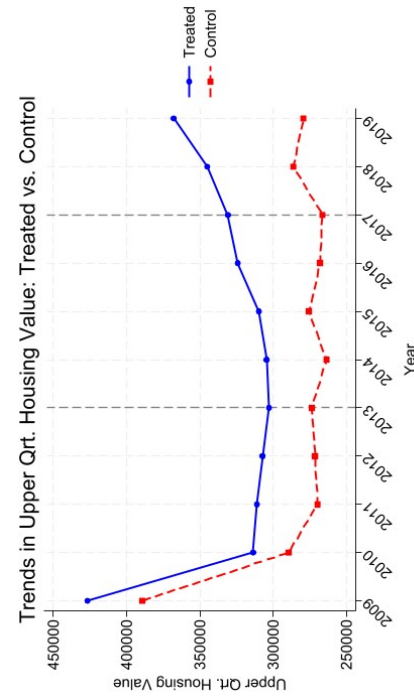
(c) Average High Qrt. Rental Prices - Levels

Figure 25: Average Rental Prices - Levels



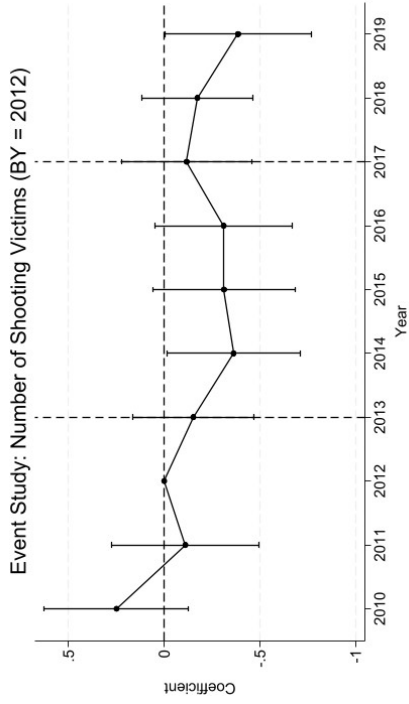
(a) Average Low Qrt. Housing Valuations - Levels

(b) Average Median Housing Valuations - Levels

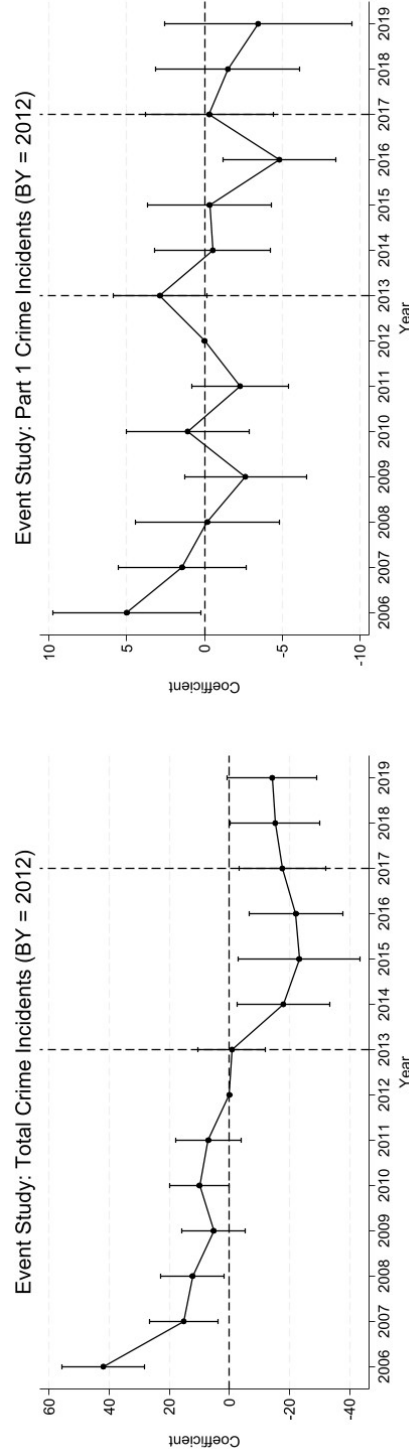


(c) Average High Qrt. Housing Valuations - Levels

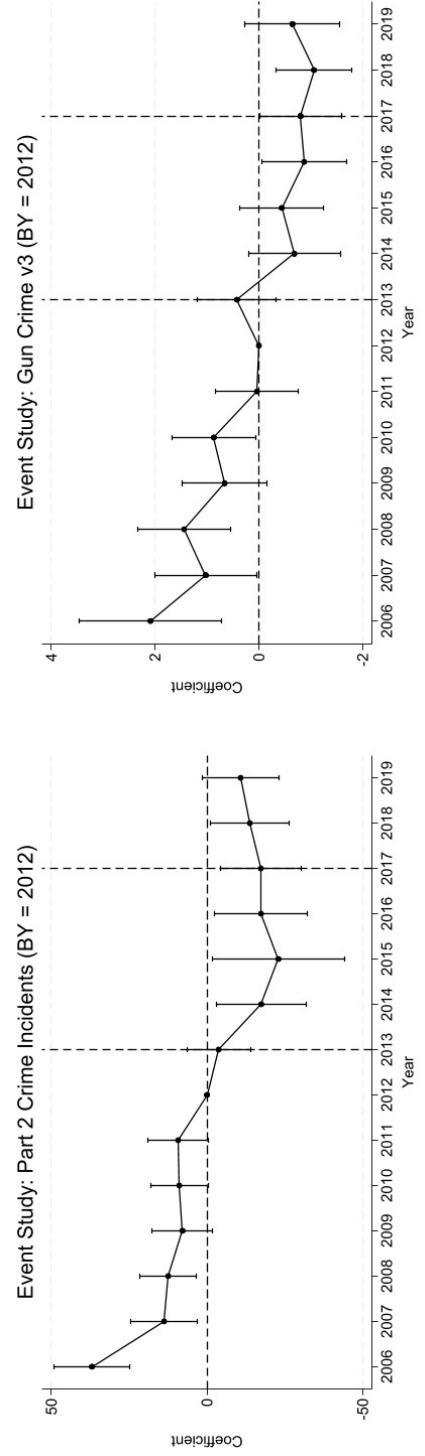
Figure 26: Average Housing Valuations - Levels



(a) PSM Levels - Event Study - Shooting Victims



(b) PSM Levels - Event Study - Total Crime Incidents

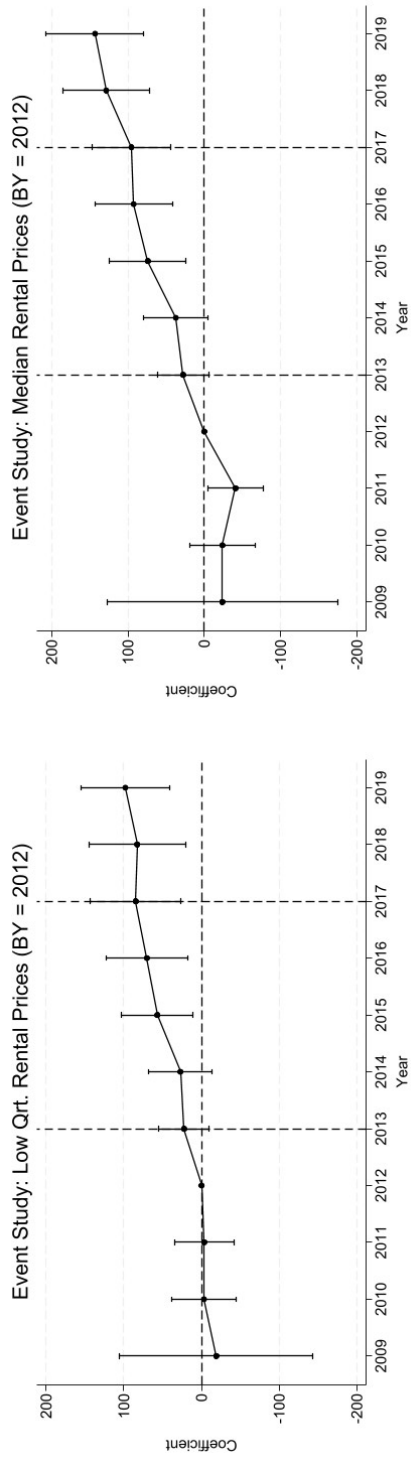


(d) PSM - Levels - Event Study - Part 2 Crime Incidents

(c) PSM Levels - Event Study - Part 1 Crime Incidents

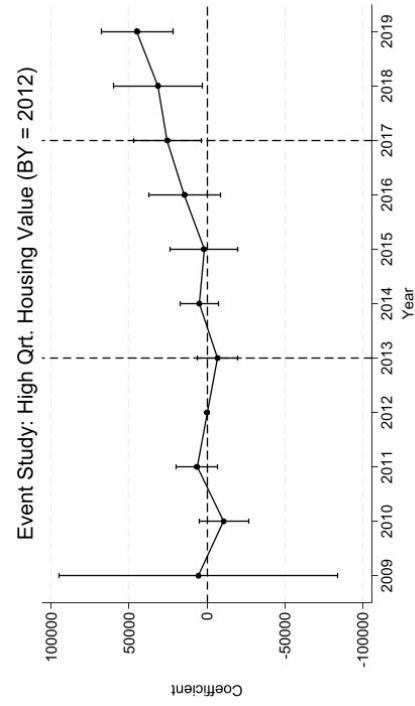
(e) PSM Levels - Event Study - Gun Crime Incidents

Figure 27: PSM Levels - Event Studies - Shooting Victims and Crime Incidents



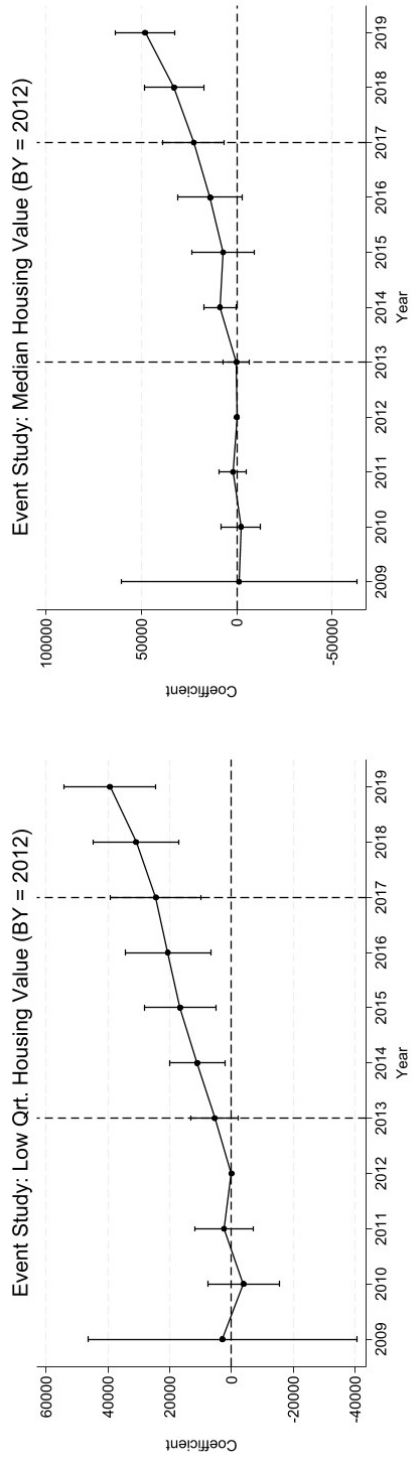
(a) PSM Levels - Event Study - Low Qrt. Rent

(b) PSM Levels - Event Study - Median Rent



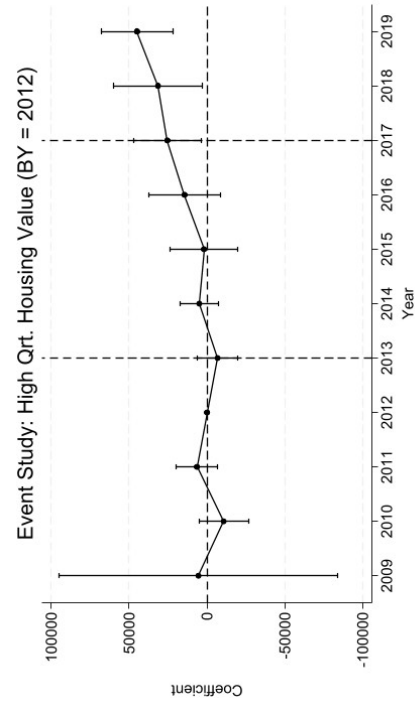
(c) PSM Levels - Event Study - High Qrt. Rent

Figure 28: PSM Levels - Event Studies - Rental Prices



(a) PSM Levels - Event Study - Low Qrt. Housing Value

(b) PSM Levels - Event Study - Median Housing Value



(c) PSM Levels- Event Study - High Qrt. Housing Value

Figure 29: PSM Level - Event Study - Housing Value

Table 14: TWFE DiD Estimates: Effects of Focused Deterrence on Crime (PSM-Level Specification)

	<i>Outcome Variable</i>				
	Shooting Victims	Total Crime	Part I Crime	Part II Crime	Gun Crime
Treat $\times$ Post	−0.297*** (0.096)	−29.460*** (6.501)	−0.972 (1.502)	−28.488*** (6.462)	−1.349*** (0.250)
Census Block Group FEs	YES	YES	YES	YES	YES
Year FEs	YES	YES	YES	YES	YES
Pre-Treatment Mean	0.917	141.340	50.636	90.703	5.509
Sample	2010–2017	2006–2017	2006–2017	2006–2017	2006–2017
Observations	2,216	3,324	3,324	3,324	3,324
Within R-squared	0.0054	0.0293	0.0004	0.0306	0.0145

Notes: TWFE OLS estimates from the propensity-score-matched sample using contemporaneous crime levels (Avg. 2009–2012) in the matching specification. Standard errors clustered at the block group level in parentheses. Sample restriction varies by outcome due to data availability. *** p<0.01, ** p<0.05, * p<0.10.

Table 15: TWFE DiD Estimates: Effects of Focused Deterrence on Rents and Housing Values (PSM-Level Specification)

	<i>Outcome Variable</i>					
	Rent			Housing Value		
	Low Qrt.	Median	High Qrt.	Low Qrt.	Median	High Qrt.
Treat $\times$ Post	67.689*** (23.202)	106.411*** (26.389)	127.947*** (35.641)	20,725*** (6,627)	19,346** (9,431)	16,005 (14,110)
Census Block Group FEs	YES	YES	YES	YES	YES	YES
Year FEs	YES	YES	YES	YES	YES	YES
Pre-Treatment Mean	691.824	878.373	1,095.593	171,264	240,774	323,391
Sample	2009–2019	2009–2019	2009–2019	2009–2019	2009–2019	2009–2019
Observations	2,695	2,766	2,785	2,935	2,970	2,949
Within R-squared	0.0078	0.0143	0.0117	0.0068	0.0031	0.0010

Notes: TWFE OLS estimates from the propensity-score-matched sample using contemporaneous crime levels (Avg. 2009–2012) in the matching specification. Standard errors clustered at the block group level in parentheses. Rents and housing valuations measured in dollars. *** p<0.01, ** p<0.05, * p<0.10.

B.3.2 Levels and Trends

Characteristic	Unmatched				Matched			
	Treated	Comp.	% Bias	<i>t</i> -stat.	Treated	Comp.	% Bias	<i>t</i> -stat.
<u>Crime Trends (2006–2012):</u>								
Part 1 Crime Count Trend	−9.466	−6.633	−14.4	−1.49	−8.521	−6.924	−8.1	−0.73
Part 2 Crime Count Trend	−41.212	−10.461	−41.4	−5.77	−33.847	−30.146	−5.0	−0.68
Gun Crime Count Trend	−4.123	−2.034	−37.4	−4.71	−3.819	−3.771	−0.9	−0.07
<u>Crime Levels (Avg. 2009–2012):</u>								
Part 1 Crime Count	49.384	52.709	−6.6	−0.64	47.760	46.715	2.1	0.27
Part 2 Crime Count	85.767	84.726	1.4	0.15	81.637	77.236	6.0	0.57
Gun Crime Count	4.582	5.683	−24.0	−2.49	4.387	4.432	−1.0	−0.08
<u>Community Characteristics (Avg. 2009–2012):</u>								
Youth Male Density	0.065	0.077	−24.0	−2.34	0.065	0.065	−0.3	−0.03
Total Population	1133.5	1160.9	−5.3	−0.58	1118.6	1128.5	−1.9	−0.17
Percent Black	0.296	0.471	−49.2	−5.34	0.296	0.268	7.8	0.70
Percent Hispanic	0.060	0.124	−43.0	−3.88	0.059	0.058	0.5	0.06
Econ. Disadvantage Index	−0.228	−0.005	−29.9	−3.22	−0.234	−0.186	−6.4	−0.58
Number of Census Block Groups	146	1,182			116	144		

Notes: Balance statistics computed via `pstest` following nearest-neighbor propensity score matching on both pre-period crime trends (2006–2012) and contemporaneous crime levels (Avg. 2009–2012). Matching procedure found the 1 nearest neighbor with a caliper of 0.03. % Bias is the standardized percentage bias in means between treated and comparison units. Rubin’s $B = 25.6$ for the matched sample. 30 treated block groups were dropped off common support.

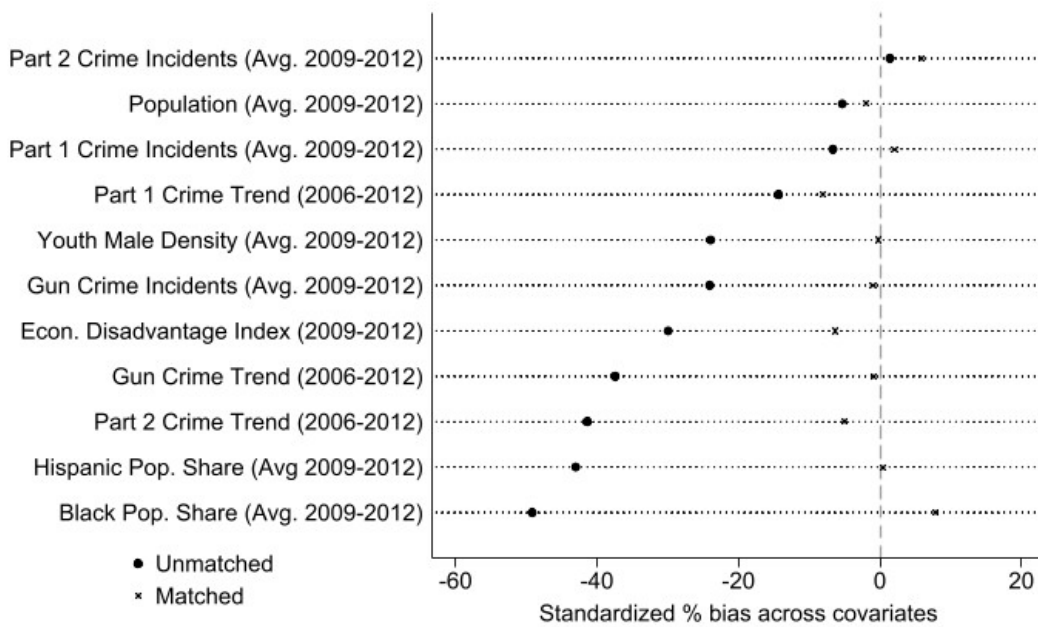


Figure 30: Covariate Balance Loveplot - Levels and Trends

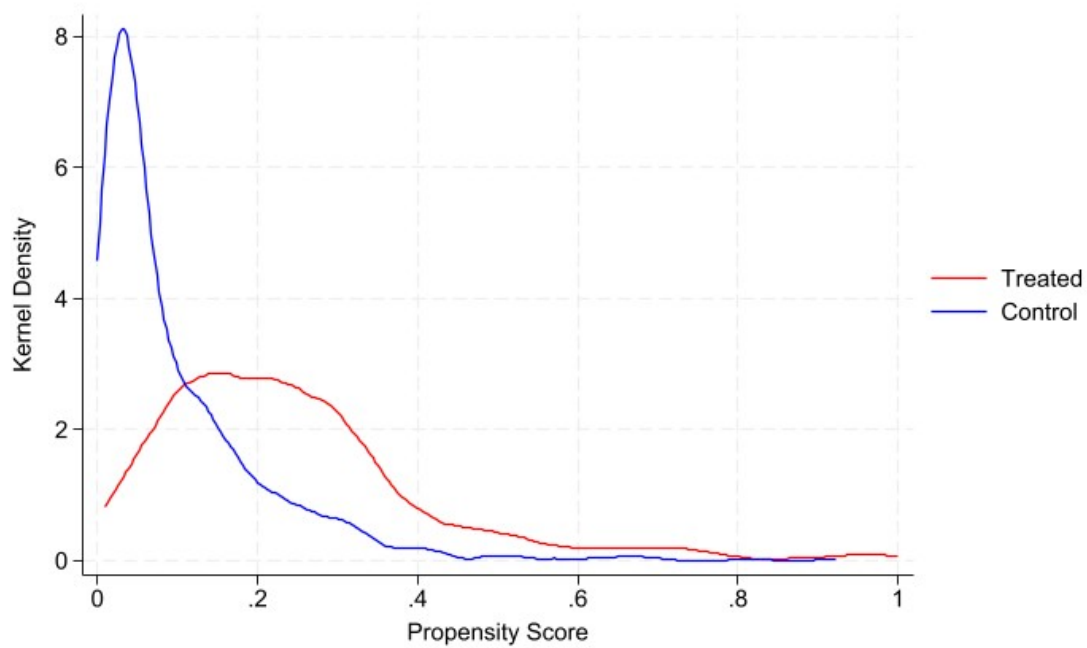


Figure 31: Kernel Density of Propensity Scores – Pre-Match - Levels and Trends

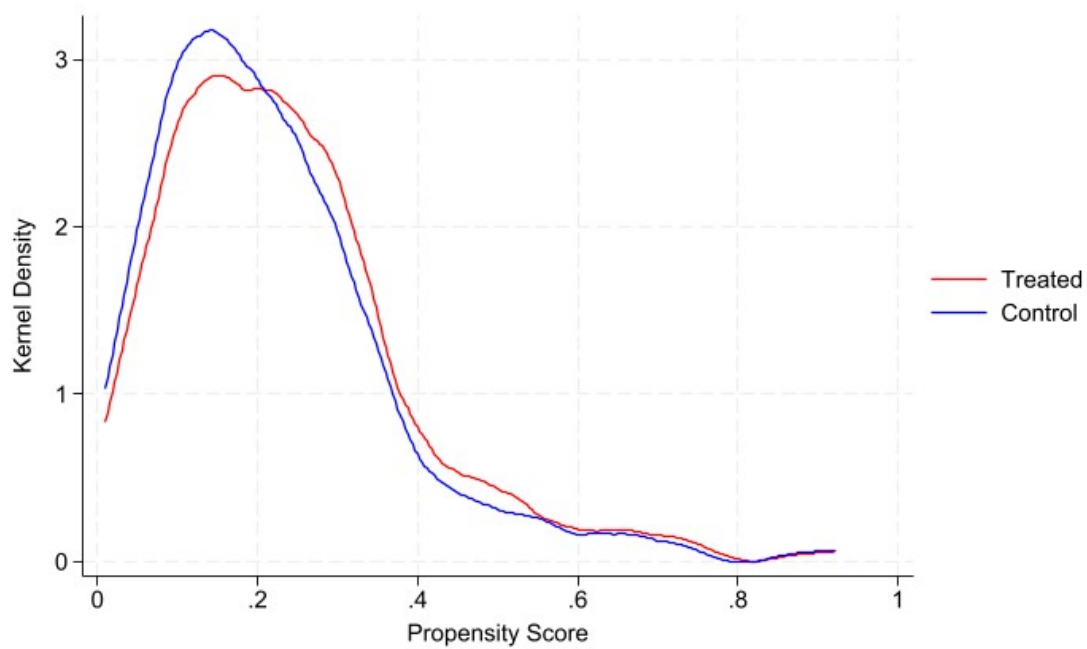
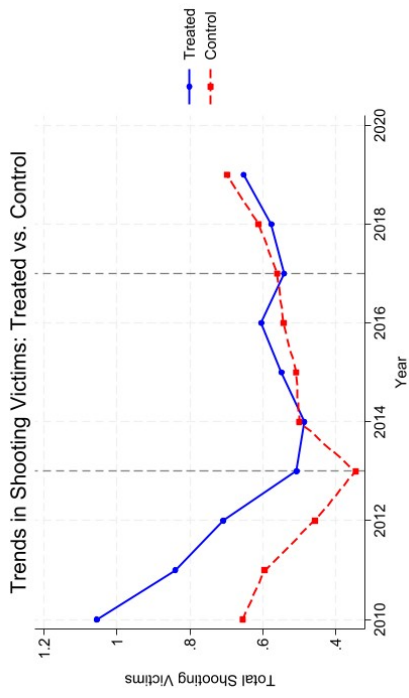
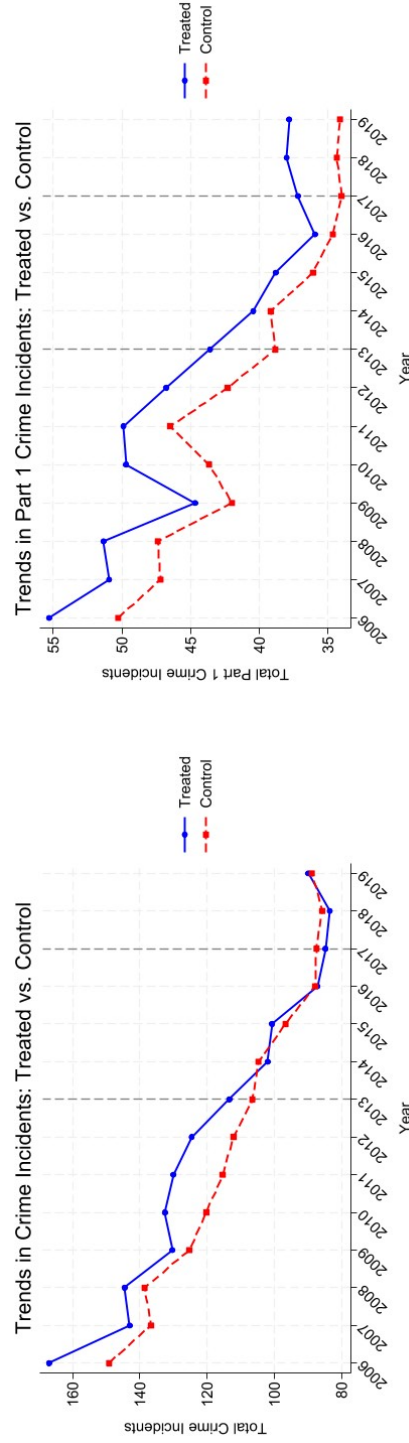


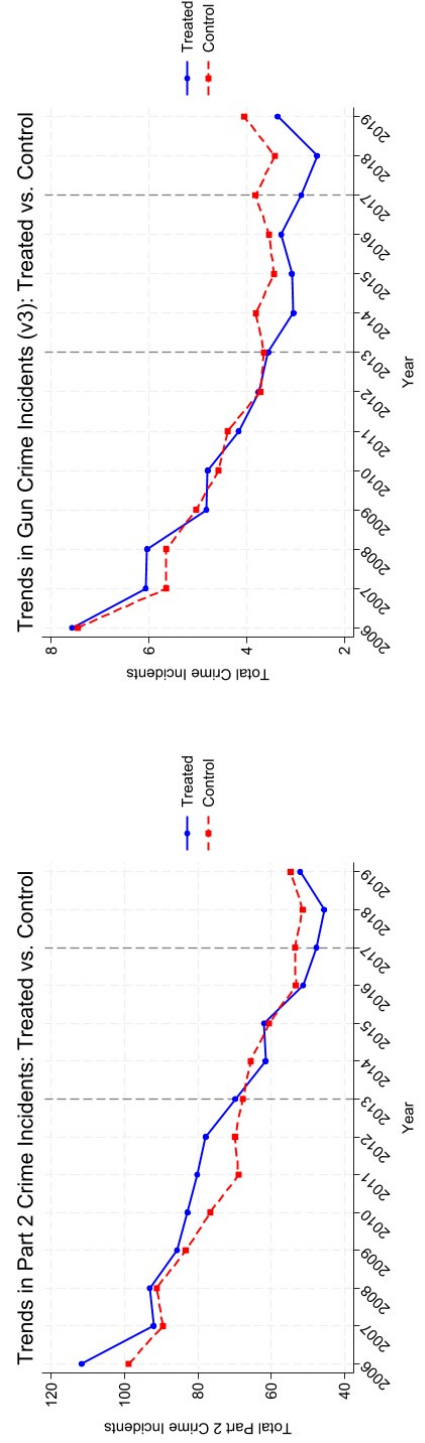
Figure 32: Kernel Density of Propensity Scores – Post-Match - Levels and Trends



(a) Average Shooting Victims (2010–2019) - Levels and Trends



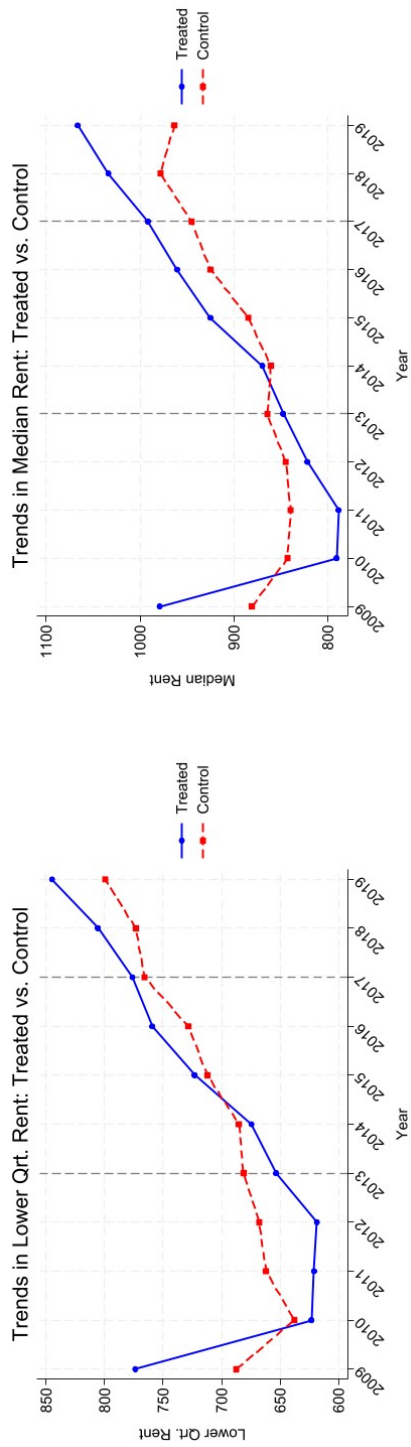
(b) Average Total Crime Incidents - Levels and Trends



(d) Average Part 2 Crime Incidents - Levels and Trends

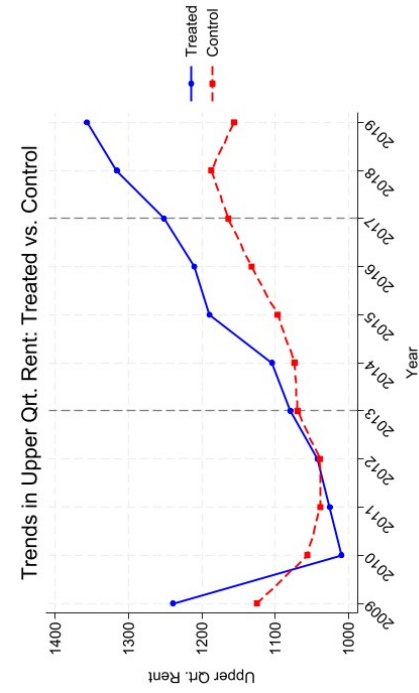
(e) Average Gun Crime (v3) Incidents - Levels and Trends

Figure 33: Post-Match Average Crime Incidents and Shooting Victims, 2006–2019 - Levels and Trends



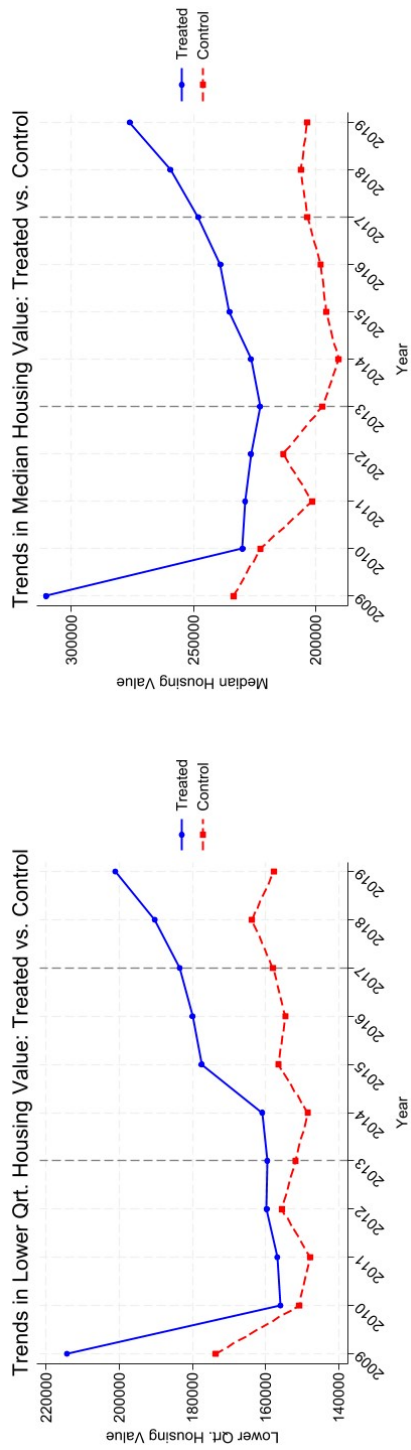
(a) Average Lower Qrt. Rental Prices - Levels and Trends

(b) Average Median Rental Prices - Levels and Trends



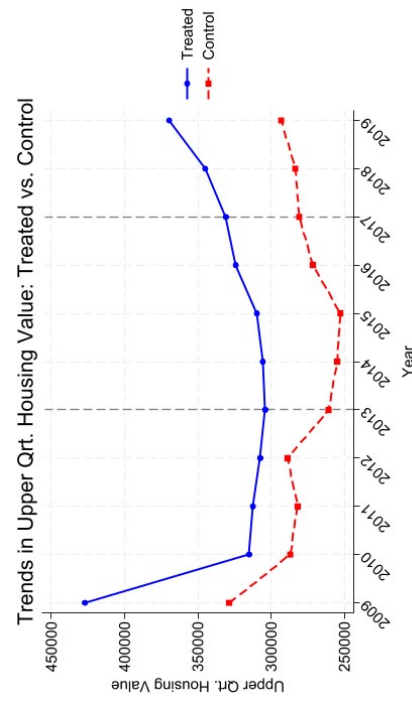
(c) Average High Qrt. Rental Prices - Levels and Trends

Figure 34: Average Rental Prices - Levels and Trends



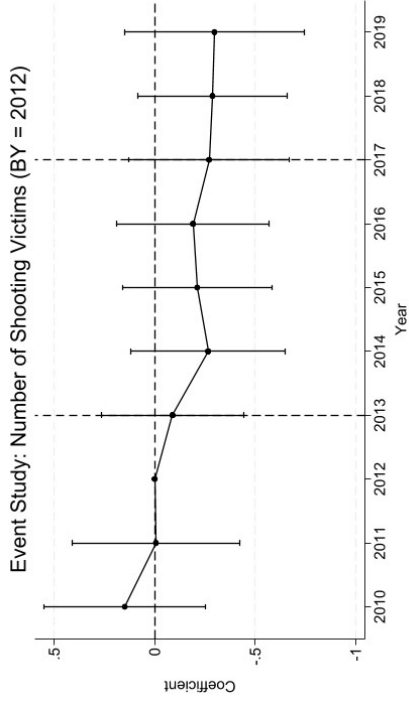
(a) Average Low Qrt. Housing Valuations - Levels and Trends

(b) Average Median Housing Valuations - Levels and Trends

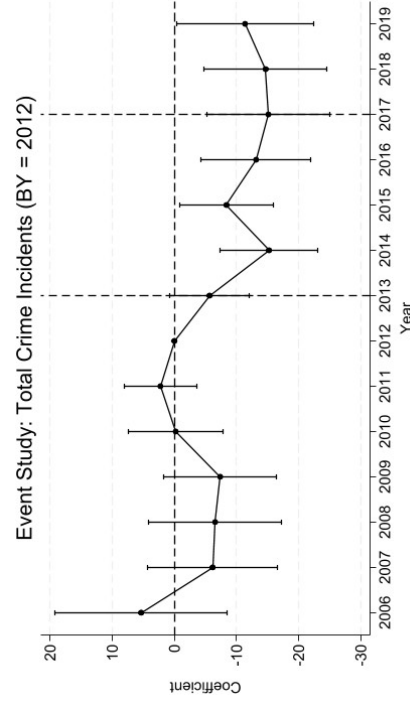


(c) Average High Qrt. Housing Valuations - Levels and Trends

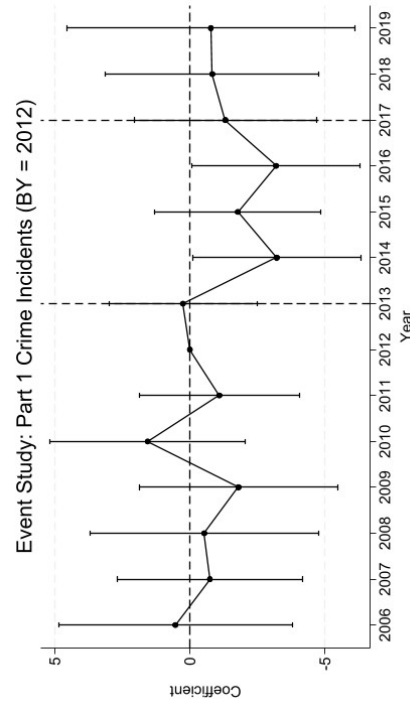
Figure 35: Average Housing Valuations - Levels and Trends



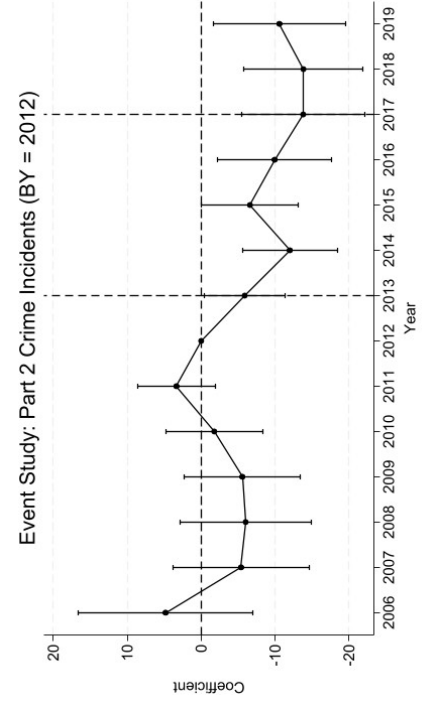
(a) PSM Levels and Trends - Event Study - Shooting Victims



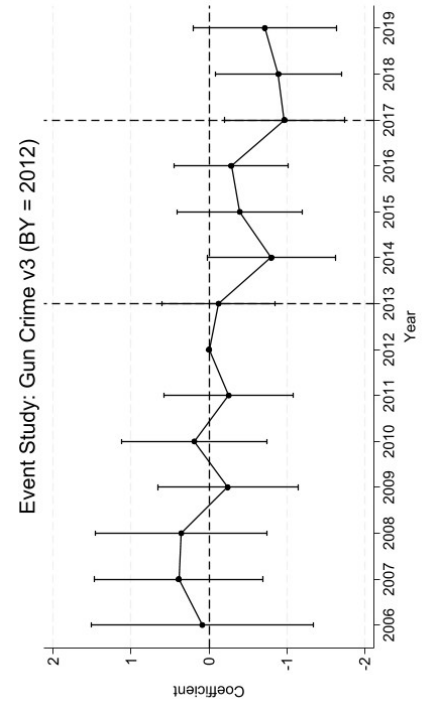
(b) PSM Levels and Trends - Event Study - Total Crime Incidents



(c) PSM Levels and Trends - Event Study - Part 1 Crime Incidents

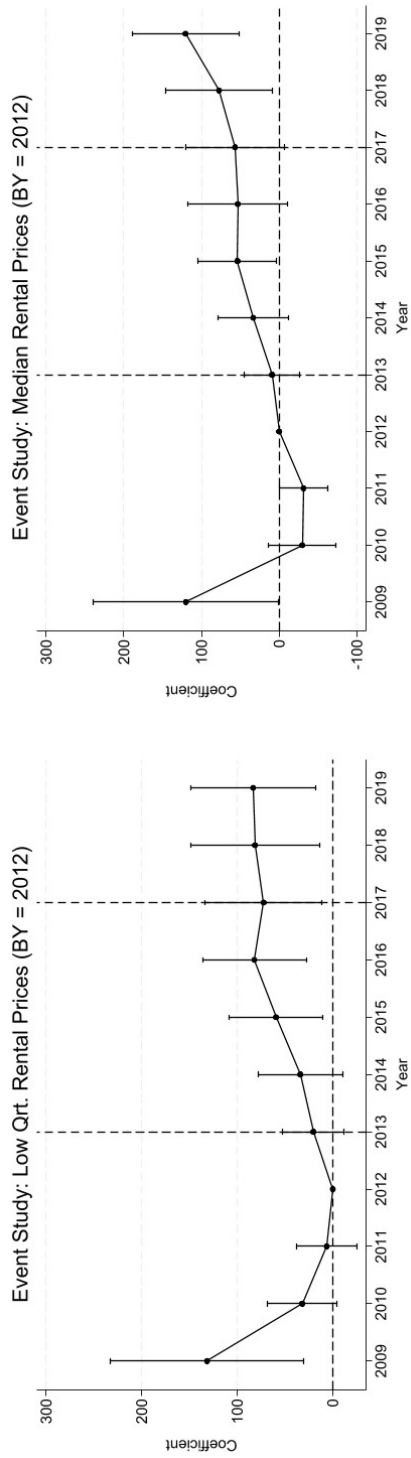


(d) PSM - Levels and Trends - Event Study - Part 2 Crime Incidents



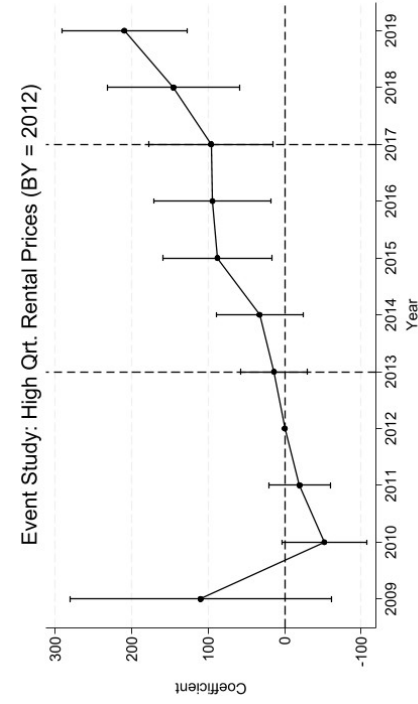
(e) PSM Levels and Trends - Event Study - Gun Crime Incidents

Figure 36: PSM Levels and Trends - Event Studies - Shooting Victims and Crime Incidents



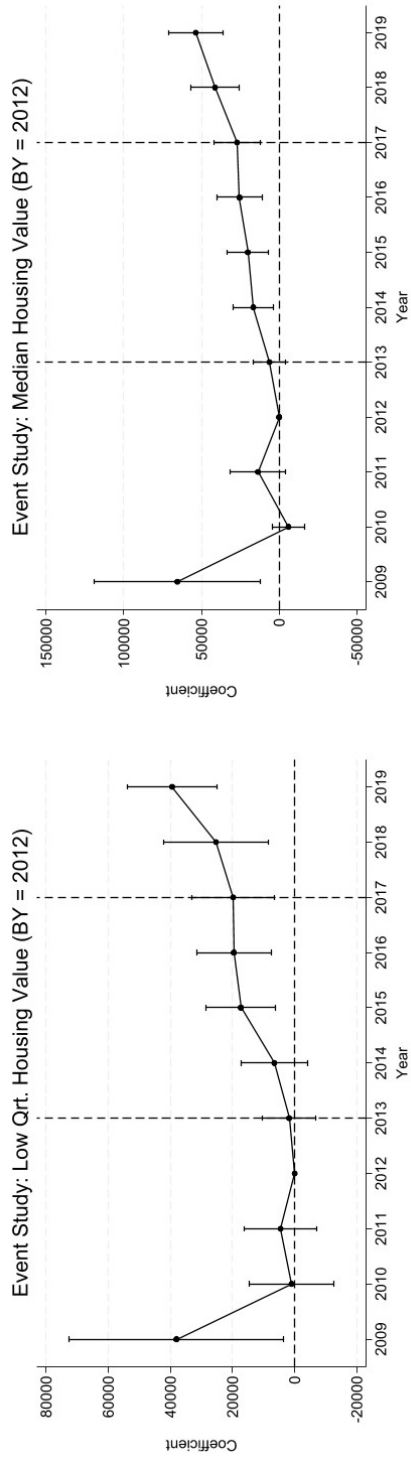
(a) PSM Levels and Trends - Event Study - Low Qrt. Rent

(b) PSM Levels and Trends - Event Study - Median Rent

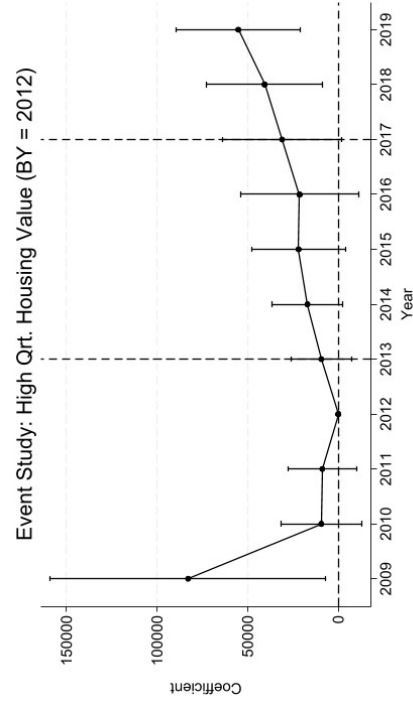


(c) PSM Levels and Trends - Event Study - High Qrt. Rent

Figure 37: PSM Levels and Trends - Event Studies - Rental Prices



(a) PSM Levels and Trends - Event Study - Low Qrt. Housing Value (b) PSM Levels and Trends - Event Study - Median Housing Value



(c) PSM Levels and Trends - Event Study - High Qrt. Housing Value

Figure 38: PSM Levels and Trends - Event Study - Housing Value

Table 16: TWFE DiD Estimates: Effects of Focused Deterrence on Crime (PSM–Levels and Trends Specification)

	<i>Outcome Variable</i>				
	Shooting Victims	Total Crime	Part I Crime	Part II Crime	Gun Crime
Treat $\times$ Post	−0.253*** (0.094)	−9.690** (4.449)	−1.553 (1.211)	−8.137** (3.738)	−0.590* (0.320)
Census Block Group FEs	YES	YES	YES	YES	YES
Year FEs	YES	YES	YES	YES	YES
Pre-Treatment Mean	0.868	138.872	49.803	89.069	5.318
Sample	2010–2017	2006–2017	2006–2017	2006–2017	2006–2017
Observations	2,080	3,120	3,120	3,120	3,120
Within R-squared	0.0044	0.0067	0.0014	0.0066	0.0028

Notes: TWFE OLS estimates from the propensity-score-matched sample using both pre-period crime trends (2006–2012) and contemporaneous crime levels (Avg. 2009–2012) in the matching specification. Standard errors clustered at the block group level in parentheses. Sample restriction varies by outcome due to data availability. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 17: TWFE DiD: Effects of Focused Deterrence on Rents and Housing Values (PSM–Levels and Trends Specification)

	<i>Outcome Variable</i>					
	Rent			Housing Value		
	Low Qrt.	Median	High Qrt.	Low Qrt.	Median	High Qrt.
Treat $\times$ Post	18.008 (23.479)	42.196 (25.674)	86.350** (35.742)	7,433 (6,379)	8,769 (8,549)	2,467 (14,797)
Census Block Group FEs	YES	YES	YES	YES	YES	YES
Year FEs	YES	YES	YES	YES	YES	YES
Pre-Treatment Mean	661.424	848.428	1,072.814	165,187	235,174	321,111
Sample	2009–2019	2009–2019	2009–2019	2009–2019	2009–2019	2009–2019
Observations	2,534	2,591	2,603	2,746	2,788	2,774
Within R-squared	0.0007	0.0027	0.0060	0.0011	0.0008	0.0000

Notes: TWFE OLS estimates from the propensity-score-matched sample using both pre-period crime trends (2006–2012) and contemporaneous crime levels (Avg. 2009–2012) in the matching specification. Standard errors clustered at the block group level in parentheses. Rents and housing valuations measured in dollars. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

C Misc. Catch All

C.1 Theoretical Framework

C.1.1 Crime as a Bundle of Complementary Activities

We adopt the standard economic view that individuals allocate time across activities to maximize expected utility [Becker \(1968\)](#); [Ehrlich \(1973\)](#). Rather than treating crime as a single binary choice, we view offending behavior as a bundle of activities chosen jointly over time. In our setting, it is useful to distinguish three broad uses of time: targeted violent group crime (e.g., shootings), other non-targeted crimes (e.g., robbery, drug-market activity, property offenses), and legal work or other non-criminal activities.

A key feature of many urban crime environments is that offense types are often complementary. The same resources, networks, and routines that support one offense can increase the payoffs to another. Group affiliation, street presence, access to firearms, and territorial control can jointly support multiple offenses. As a result, individuals involved in shootings are frequently the same individuals involved in other crimes, and changes in incentives for one type of crime may shift behavior along multiple offense margins at once.

At the same time, crime categories can also be substitutes. When the expected cost of one offense rises relative to another, individuals may reallocate effort toward alternative offenses that are perceived as less risky or less exposed to enforcement. Whether complementarity or substitution dominates is theoretically ambiguous and depends on how closely different offenses share inputs, organization, and exposure to sanctions. This ambiguity about the particular effects of targeted crime reduction causes the identification of any policy ramification to be an empirical question.

C.1.2 Focused Deterrence as a Targeted Shock to Incentives

Focused deterrence strategies, including Philadelphia’s implementation, target violent groups and their most serious offenses. The intervention is designed to increase both the perceived certainty and the perceived severity of punishment associated with targeted violence. Call-ins aim to increase the salience and credibility of enforcement by communicating a clear deterrence message and specifying

the consequences of continued violence. Lever-pulling operationalizes that message through coordinated actions across agencies, including prosecutions, probation and parole responses, warrant enforcement, and other sanctions [Braga et al. \(2017, 2018\)](#).

Because focused deterrence is organized at the group level, it can affect behavior through three related channels. First, it directly changes incentives for the targeted violent offenses. Second, by raising the expected costs of participating in the group-based platform that supports violence, it can indirectly affect other offenses that are complementary to targeted violence. If shootings and other crimes share people, resources, and organization, then reducing targeted violence may also reduce related non-targeted crimes. Third, the intervention can operate through broader reputational and informal mechanisms. For example, coordinated enforcement can increase perceived surveillance, raise the reputational costs of offending, and change expectations about community tolerance and system-wide attention, which can influence behavior even beyond the narrow set of targeted acts.

At the same time, focused deterrence may also induce substitution. If some individuals can shift away from targeted violence toward offenses that are less tied to group enforcement levers or less salient to coordinated responses, non-targeted crime can increase in certain forms even as targeted violence falls. Thus, theory allows for both positive spillovers (through complementarity and platform effects) and offsetting substitution, with the net response depending on local criminal organization and the scope of enforcement.

C.1.3 From Crime Reductions to Housing and Local Economic Activity

The final step connects changes in neighborhood crime conditions to housing markets and local economic activity. In a Rosen–Roback spatial equilibrium, local amenities and disamenities are capitalized into housing values and rents [Rosen \(1974\)](#); [Roback \(1982\)](#). Crime risk is a salient neighborhood disamenity. When crime falls, neighborhoods become more attractive to residents and firms, which can raise property values, increase rents through rental-market adjustment, and encourage commercial activity such as new business entry.

In our context, the relevant change is the overall shift in neighborhood safety induced by focused deterrence, including any spillovers from targeted violence to other crimes through complementarity

and platform effects. Improvements in safety can therefore translate into changes in housing-market outcomes and local economic activity through standard capitalization and location-choice mechanisms.